▶ Notebook

Lecture 5 at a glance: six parts, twenty-five sections

Part 1 · Why generate?

1.1 Things that do not exist ▶
1.2 Text, pictures and sound ▶
1.3 What p(x) means, and what it is for ▶
1.4 What they all share ▶

Part 2 · Autoencoders: squeeze, then rebuild

2.1 An image is 784 numbers ▶
2.2 Encoder, code, decoder ▶
2.3 The loss, and training ▶
2.4 What the code holds ▶

Part 3 · Can an autoencoder generate?

3.1 Decode random codes ▶
3.2 Where the codes are ▶
3.3 What we want instead ▶

Part 4 · From an autoencoder to a VAE: three fixes

4.1 Fix 1: choose where codes live ▶
4.2 Fix 2: a cloud for each image ▶
4.3 Fix 3: pull the clouds to N(0, I) ▶
4.4 The loss the fixes give ▶

Part 5 · Why this loss: Lecture 1's ELBO, one image at a time

5.1 Same problem as Lecture 1 ▶
5.2 A model with a hidden code ▶
5.3 The same wall as Lecture 1 ▶
5.4 The variational approximation ▶
5.5 The loss, derived from the ELBO ▶
5.6 Both terms, live ▶
5.7 What the VAE learned ▶
5.8 The KL weight β ▶

Part 6 · Putting it together

6.1 The whole lecture in maths ▶
6.2 Wrap-up ▶

> Six parts, twenty-five numbered sections, about 92 minutes. The ▶ Notebook link on every slide opens its section in Colab.

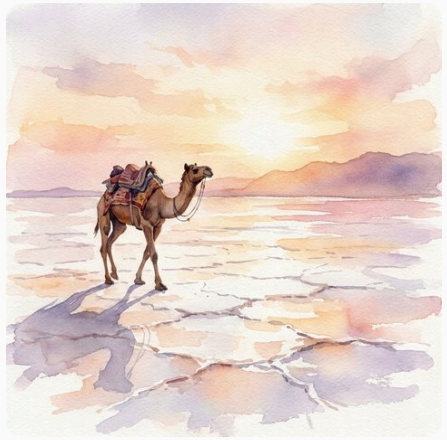
▶ Notebook

Part 1 · Why generate?

1 · Why generate? | 2 · Autoencoders | 3 · Can it generate? | 4 · Three fixes | 5 · Why: the ELBO | 6 · Putting it together



faces (StyleGAN)



text to image



a map to a landscape



digits: today

1.1 Things that do not exist ▶

1.2 Text, pictures and sound ▶

1.3 What p(x) means, and what it is for ▶

1.4 What they all share ▶

> Generative models make new text, pictures and sound. A quick tour, what they are good for, and the one recipe they all share.

▶ Notebook 1.1

Who are these people?



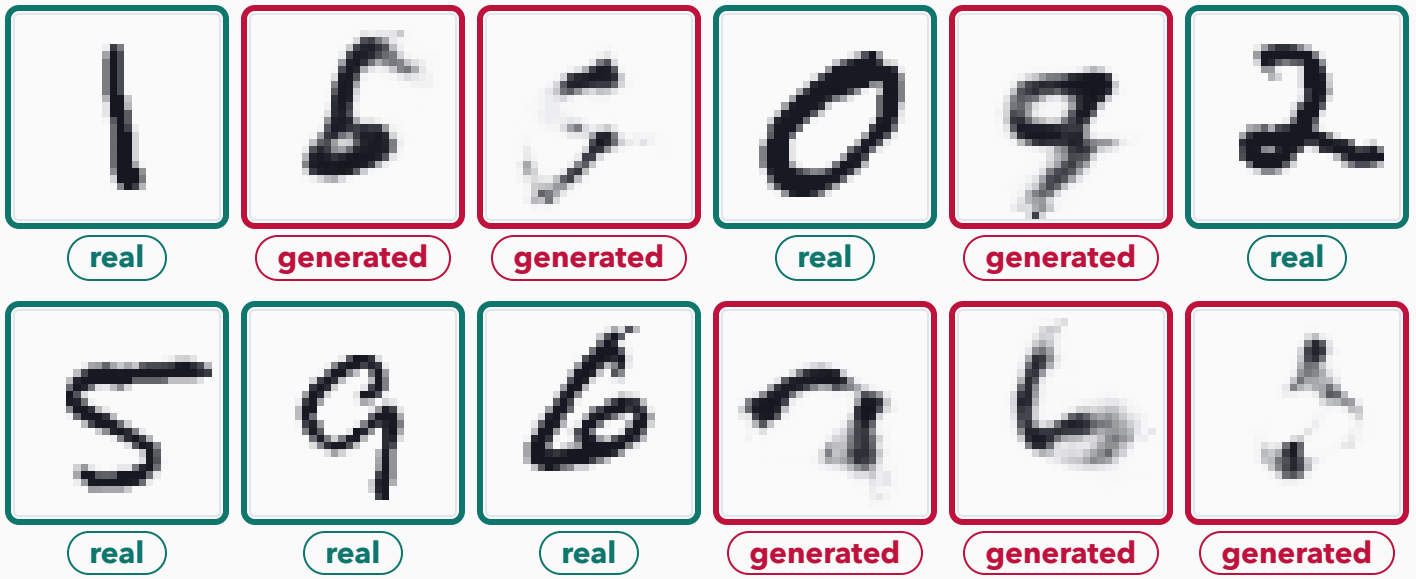
None of them exist. A network turned random numbers into each face.

> Nobody. Each face is a sample from a generative model (StyleGAN, 2019): random numbers in, a face out.

Faces generated by StyleGAN: Karras, Laine and Aila, "A Style-Based Generator Architecture for GANs", CVPR 2019. NVIDIA, CC BY-NC 4.0.

▶ Notebook 1.1

Real or generated? Six of these twelve digits are fake



Six are handwritten MNIST test digits. Six come from a network.

The fakes: numbers 2, 3, 5, 10, 11, 12, counting left to right, top to bottom.

By the end of today you will know how the network that made them was trained: a VAE with 16 numbers per code.

> The six outlined in red are generated. Some give themselves away with broken strokes; some fool most people.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 1.2

A chatbot is a generative model: one prompt, three different answers

Prompt: Write one sentence that opens a short story about the first monsoon rain in Gandhinagar.

Sample 1. As the first heavy drops of the monsoon struck the sunbaked avenues of Sector 21, the parched earth of Gandhinagar exhaled a sweet, dusty petrichor that signaled the end of the long summer.

Sample 2. As the first heavy drops struck the sun-baked asphalt of Sector 17, Gandhinagar exhaled its scorching summer heat in a sweet, petrichor-laced sigh that rippled down the quiet, tree-lined avenues.

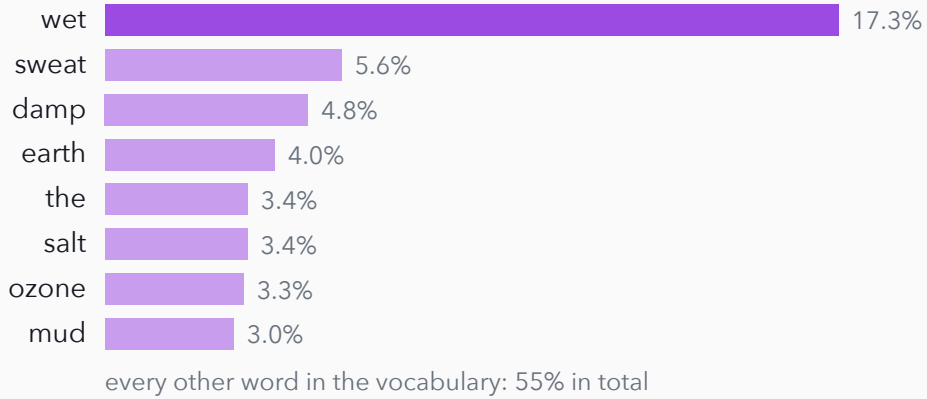
Sample 3. The brutal summer heat surrendered in a single afternoon when heavy, bruised clouds cracked open over Gandhinagar, washing its dusty neem trees and quiet, grid-lined sectors in the intoxicating scent of the season's first rain.

Same prompt, three different replies. Each is a sample from a distribution over sentences.

▶ Notebook 1.2

Under the hood: a probability for every possible next word

“After months of heat, the first monsoon rain smelled of …”



Sampling = pick a word with these probabilities, append it, repeat.

The same model’s three continuations: “melons and oranges.”, “sweat and damp ruins. It was moist, it was s”, “desperation. What had occurred, and what transpired that its”.

> Sample a word, append it, repeat. That is all a chatbot does when it writes, at a much larger scale.

SmolLM2-360M-Instruct (Hugging Face), next-token probabilities and three samples at temperature 1, computed for Lecture 2.

▶ Notebook 1.2

Text to image: one prompt, four samples

Prompt: A watercolour painting of a camel walking across the white salt desert of the Rann of Kutch at sunrise, soft light, wide view.



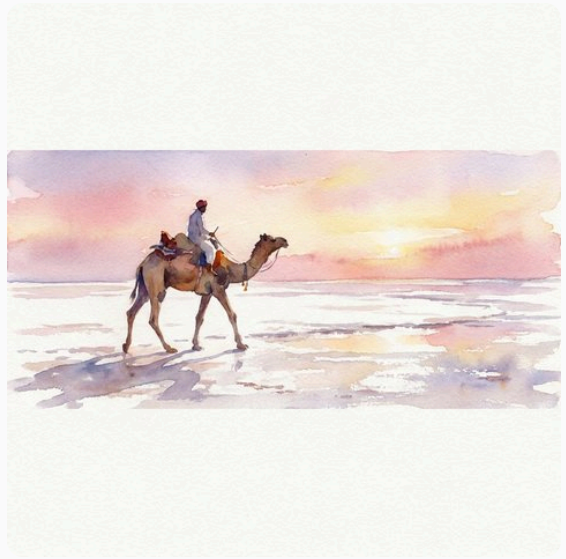
sample 1



sample 2



sample 3



sample 4

Three more runs, same prompt: a rider appears, a walker, a different horizon. Four samples.

▶ Notebook 1.2

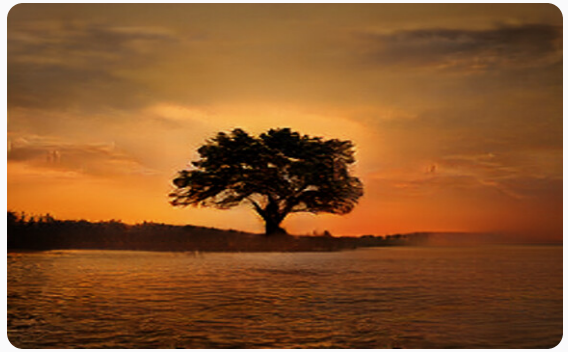
Image in, image out



one label map



sample 1



sample 2



sample 3

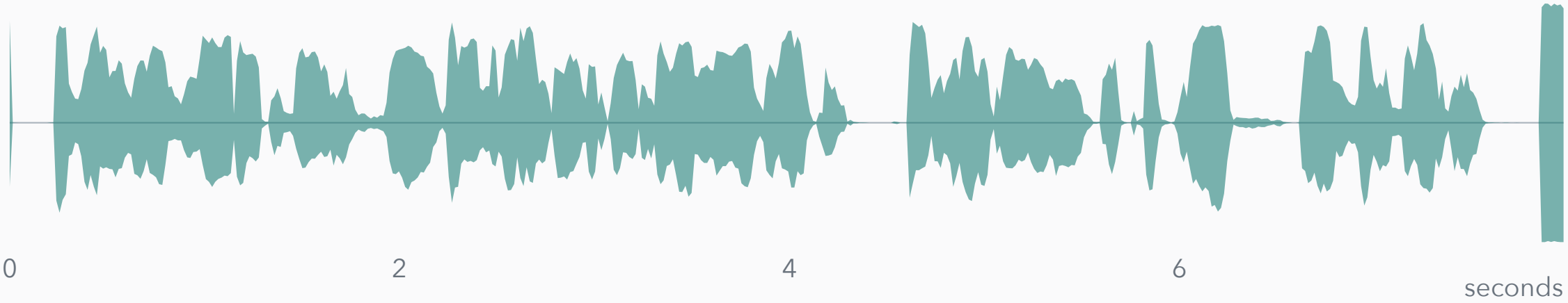
> The same label map, sampled three times: three different photographs. Still a distribution, now given an input.

Isola et al., pix2pix (CVPR 2017); Park et al., SPADE (CVPR 2019); Zhu et al., CycleGAN (ICCV 2017). Attributed teaching excerpts; copyright with the authors.

▶ Notebook 1.2

Sound: text in, 24,000 numbers per second out

Text: “A generative model turns a few random numbers into a new example: a sentence, a picture, or this voice.”



0:00 / 0:00

191,219 numbers for 8.0 seconds. A few of them: 0.102, 0.089, 0.083, 0.072, 0.062, 0.051, …

The output is just a long list of numbers: the air pressure, 24,000 times a second. Generating sound = generating that list.

▶ Notebook 1.3

First, the names: x is the data, y is a label

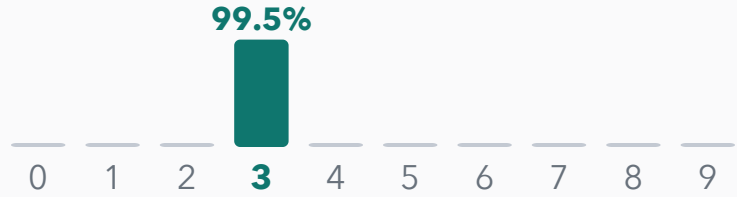
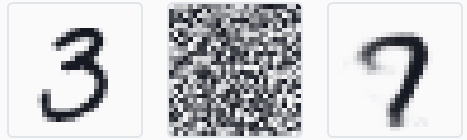
x = the data



one image: 784 numbers, one per pixel: (…, 0.00, 0.64, 0.99, 0.99, 0.99, …)

y = a label about x

3 which digit: one of 0, 1, …, 9

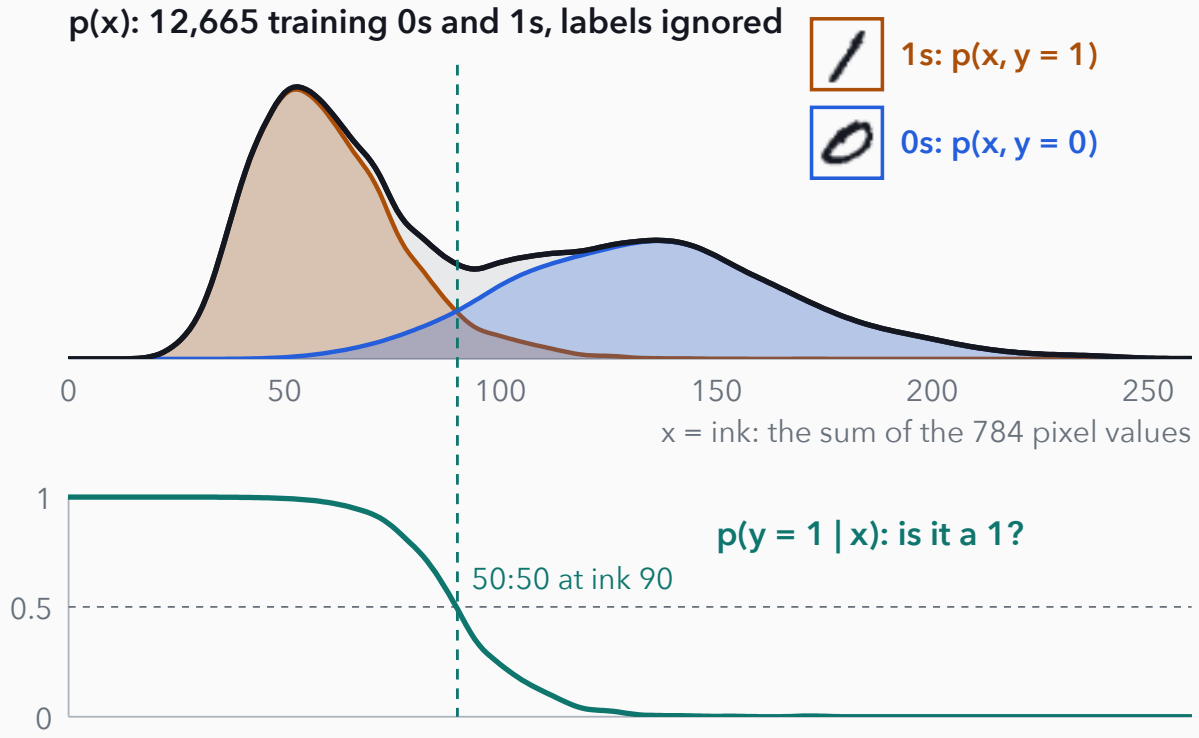
$p(y \mid x)$	Given this image, how probable is each label? "|" reads "given". **discriminative: a classifier**	
$p(x)$	How probable is this image itself, with no label? And: draw a new one. **generative: today**	 high almost 0 a new x
$p(x \mid y{=}3)$	Fix the label, then ask: which images? Every way to write a 3. **conditional generative**	

> p(x | y = 3): fix the label first, then ask which images are likely: every way to write a 3. The real 3s are draws from it. A text-to-image model learns p(image | prompt), with the prompt as y.

Classifier and numbers: `lecture5/scripts/xy.py`. Digits from Lecture 2's data; the new x is a draw from its 16-number VAE.

▶ Notebook 1.3

Generative or discriminative, in one picture



To draw it: one number per image, its ink. Only the 0s and 1s.

$$p(x, y) = p(y)\, p(x \mid y)$$

$$p(x) = p(x, 0) + p(x, 1)$$

$$p(y \mid x) = \frac{p(x, y)}{p(x)}$$

Generative learns the top: it can score and draw x, and the bottom follows.
Discriminative learns only the bottom: it can label, not draw.

Divide the 1s' hump by the total and you get p(y = 1 | x): near 1 for little ink, near 0 for a lot, 50:50 at ink 90. A classifier learns only this curve. A generative model learns the humps, and gets this curve free by Bayes' rule.

Numbers: `lecture5/scripts/xy.py` (MNIST training set).

▶ Notebook 1.3

Beyond pictures: generative models in science

Weather

GenCast samples 50 possible futures for the next 15 days. The spread of the samples says how sure the forecast is.

Price et al., Nature 2025

Proteins

RFdiffusion generates new protein shapes, for example ones that bind a chosen target.

Watson et al., Nature 2023

Materials

MatterGen proposes new crystal structures with properties you ask for.

Zeni et al., Nature 2025

Missing data

Sensors drop readings. A model of p(x) fills the gap with plausible values instead of a flat average, and says how unsure it is.

for example, air-quality and smart-meter records

> Materials, and the everyday case: filling in missing sensor readings with plausible values.

Price et al., "Probabilistic weather forecasting with machine learning" (GenCast), Nature 2025; Watson et al., "De novo design of protein structure and function with RFdiffusion", Nature 2023; Zeni et al., "A generative model for inorganic materials design" (MatterGen), Nature 2025.

▶ Notebook 1.3

What a model of p(x) is good for

Create

new text, images, sound, designs

Fill in

missing pixels, missing readings, the rest of a sentence

Simulate

many plausible futures instead of one average

Spot the odd one out

a reading with very low p(x) is suspicious: a faulty sensor, a fraud

Compress

likely data needs fewer bits: Lecture 1's codes

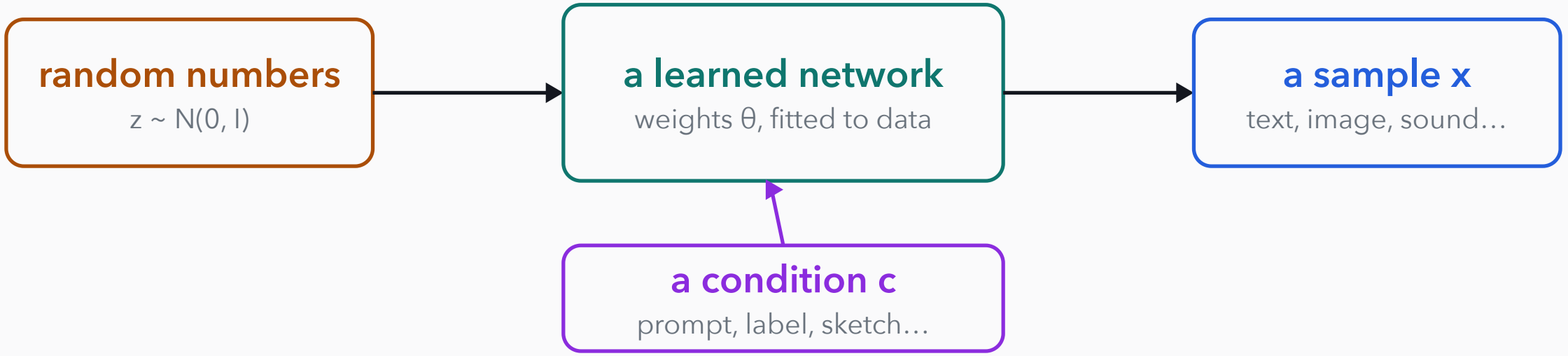
Make more data

extra training examples, or shareable synthetic records

> Three that need scores: spot anomalies, compress, and generate extra data when real data is scarce or private.

▶ Notebook 1.4

What they share: random numbers in, a learned network, a sample out



> Most also take a condition c that steers the sample: a prompt, a class label, a sketch. Today we drop c and learn p(x) for digits.

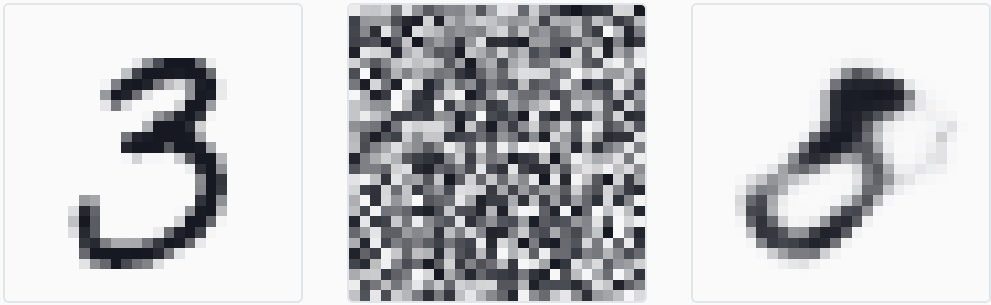
▶ Notebook 1.4

A model of p(x) can do two jobs: score and sample

Score: how plausible is this x?

$$p(\text{a clean } 3) \text{ high}, \qquad p(\text{static}) \text{ tiny}$$

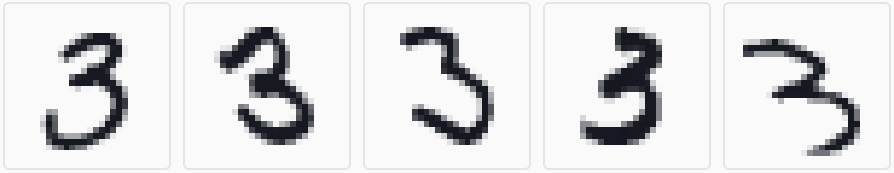
Sample: make a new x that looks like the data.



> Job two, sample: produce a new x. Both jobs come from the same fitted distribution.

▶ Notebook 1.4

Why not just output the average?



average of 100 threes:



Nobody writes this 3. The average is the blur in the middle of many real 3s.

> The average looks like no real 3. We need the whole distribution, so we can draw one sharp 3 at a time.

▶ Notebook 1.4

Today: two networks, then Lecture 1's objective

Part 2 · Autoencoder

One network squeezes a digit into 2 numbers; a second rebuilds it.

Part 3 · Can it generate?

Feed the second network random numbers. What comes out, and why?

Part 4 · Three fixes

Codes from N(0, I), a cloud per image, a KL pull: a new loss.

Part 5 · Why this loss

A model with a hidden code: the loss is minus Lecture 1's ELBO.

Part 6 · Together

Every term of the loss traced back to KL and the ELBO.

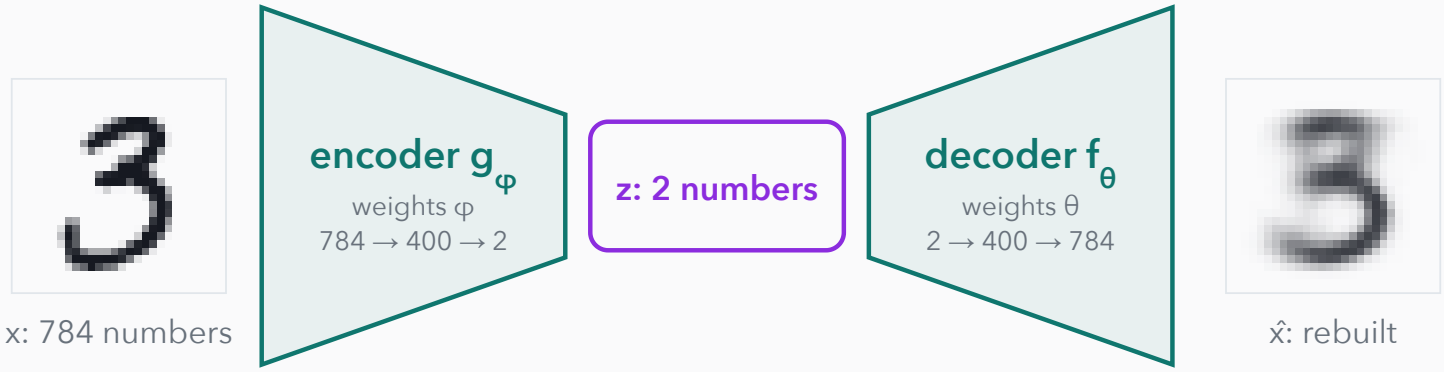
The VAE's loss for one image:

$$\underbrace{\frac{1}{2\sigma^2}\left\|x-\hat{x}\right\|^2}_{\text{rebuild it}}+\underbrace{\mathrm{KL}\left(q_\phi(z\mid x)\,\|\,\mathcal{N}(0,I)\right)}_{\text{Lecture 1's KL to the prior}}$$

> A last part traces every term of that loss back to KL and the ELBO.

▶ Notebook

Part 2 · Autoencoders: squeeze, then rebuild



2.1 An image is 784 numbers ▶
2.2 Encoder, code, decoder ▶
2.3 The loss, and training ▶
2.4 What the code holds ▶

> Squeeze each digit into 2 numbers, then rebuild it from them. Two small networks, trained together on one squared-error loss.

▶ Notebook 2.1

MNIST: 70,000 handwritten digits, 28 × 28 pixels each



0	0	0	0.3	0.8	1.0	1.0	1.0	1.0	1.0	1.0
0	0.0	0.5	1.0	1.0	0.9	0.6	0.3	0.2	0.5	1.0
0	0.4	1.0	0.9	0.6	0.1	0	0	0.0	0.5	1.0
0	0.1	0.6	0.1	0	0	0	0	0.2	1.0	1.0
0	0	0	0	0	0.0	0.4	0.6	0.9	1.0	0.9
0	0	0	0	0	0.3	1.0	1.0	1.0	0.9	0.2
0	0	0	0.6	0.9	1.0	1.0	1.0	1.0	1.0	0.4
0	0	0	0.9	1.0	1.0	1.0	0.7	1.0	1.0	1.0
0	0	0	0.0	0.2	0.0	0.0	0	0.2	0.7	1.0
0	0	0	0	0	0	0	0	0	0.1	0.9

60,000 for training, 10,000 for testing.

Our running example: test digit #90, a 3. Each pixel is a number: 0 is paper, 1 is full ink.

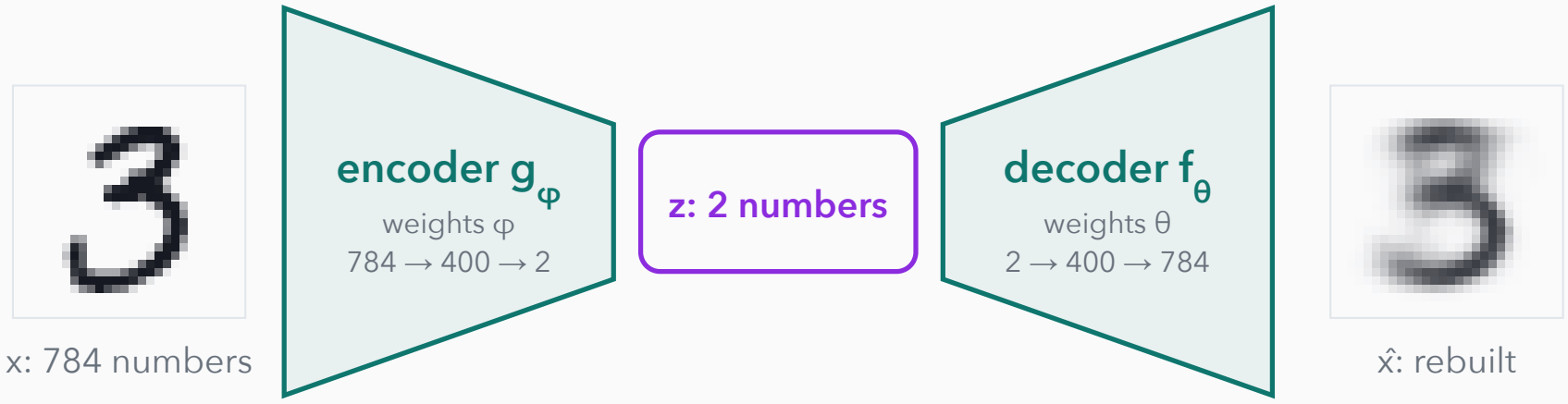
$$x \in [0, 1]^{784}$$

> We follow one digit all hour: test digit #90, a 3. To a network it is a list of 784 numbers between 0 and 1.

MNIST (LeCun, Cortes and Burges), grey levels scaled to [0, 1].

▶ Notebook 2.2

An autoencoder: encoder, short code, decoder



The decoder f_θ, with weights θ, mirrors it: 2 → 400 → 784, then a sigmoid. Train both so that $\hat{x} \approx x$.

▶ Notebook 2.2

Two small networks in PyTorch

```python
class AutoEncoder(nn.Module):
    def __init__(self, z=2):
        super().__init__()
        self.encoder = nn.Sequential(  # g_φ, weights φ
            nn.Linear(784, 400), nn.ReLU(),
            nn.Linear(400, z))
        self.decoder = nn.Sequential(  # f_θ, weights θ
            nn.Linear(z, 400), nn.ReLU(),
            nn.Linear(400, 784), nn.Sigmoid())

    def forward(self, x):
        z = self.encoder(x)     # 784 -> z numbers
        return self.decoder(z)  # z -> 784 numbers
```

Why a sigmoid at the end? Are we classifying?

No: it keeps every output pixel between 0 and 1, like the data.

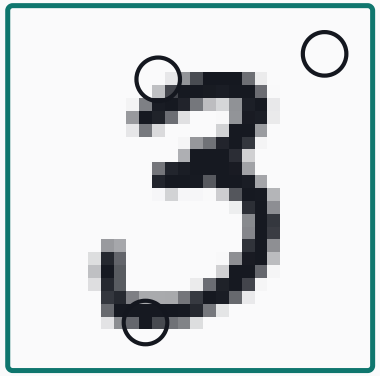
> The final sigmoid squashes each of the 784 outputs into (0, 1), the range of our pixels.

▶ Notebook 2.3

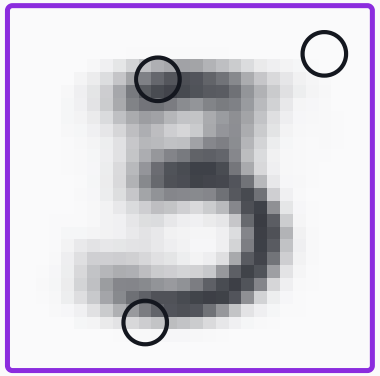
The loss: how far each rebuilt pixel is from the real one

$$L(\phi,\theta) = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{784}\left(x_{ij} - \hat{x}_{ij}\right)^2, \qquad \hat{x}_i = f_\theta\left(g_\phi(x_i)\right)$$

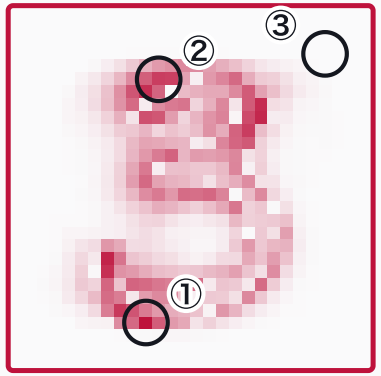
target x: our 3



prediction $\hat{x}$



$(x - \hat{x})^2$, each pixel



sum of all 784 = 26.3

pixel	x	$\hat{x}$	$(x - \hat{x})^2$
① missed ink	0.99	0.20	0.63
② extra ink	0.00	0.65	0.42
③ background	0.00	0.00	0.00
all 784 pixels			**26.3**

> Three pixels: ① missed ink costs 0.63, ② ink where the 3 has none costs 0.42, and ③ clean background costs 0.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

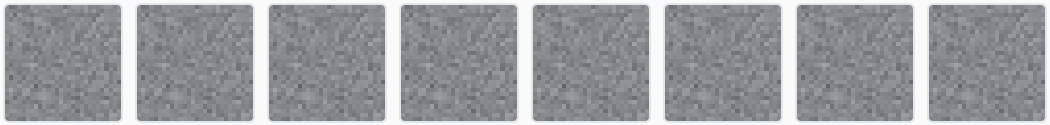
▶ Notebook 2.3

Before training: always look at what the network outputs

8 test digits



untrained output



Every output pixel is between 0.38 and 0.66: flat grey, whatever the input.

Each pixel is off by about 0.5, since the target is 0 or 1. Squared: about 0.24.

$$\underbrace{784}_{\text{pixels}} \times \underbrace{0.235}_{\text{per pixel}} \approx 185$$

> Each of 784 pixels is off by about a half, which squares to a quarter: about 185 in all. Training must bring this far down.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 2.3

One training step: forward, loss, backward, update

```
model = AutoEncoder(z=2)
opt = torch.optim.Adam(model.parameters(), lr=1e-3)
for epoch in range(30):
    for x in loader:          # 128 images at a time
        x_hat = model(x)
        loss = ((x_hat - x)**2).sum(1).mean()
        opt.zero_grad()
        loss.backward()       # encoder AND decoder
        opt.step()
```

469 steps per pass over the 60,000 digits; 30 passes.

One loss, one backward pass: the gradient flows from the pixel errors back through the decoder, the code, and the encoder.

> loss.backward() sends gradients through the decoder, into the code, and on into the encoder: both are trained together.

▶ Notebook 2.3

Press play: reconstructions sharpen, then stop improving

8 test digits



what it outputs after 469,000 steps (1,000 epochs)

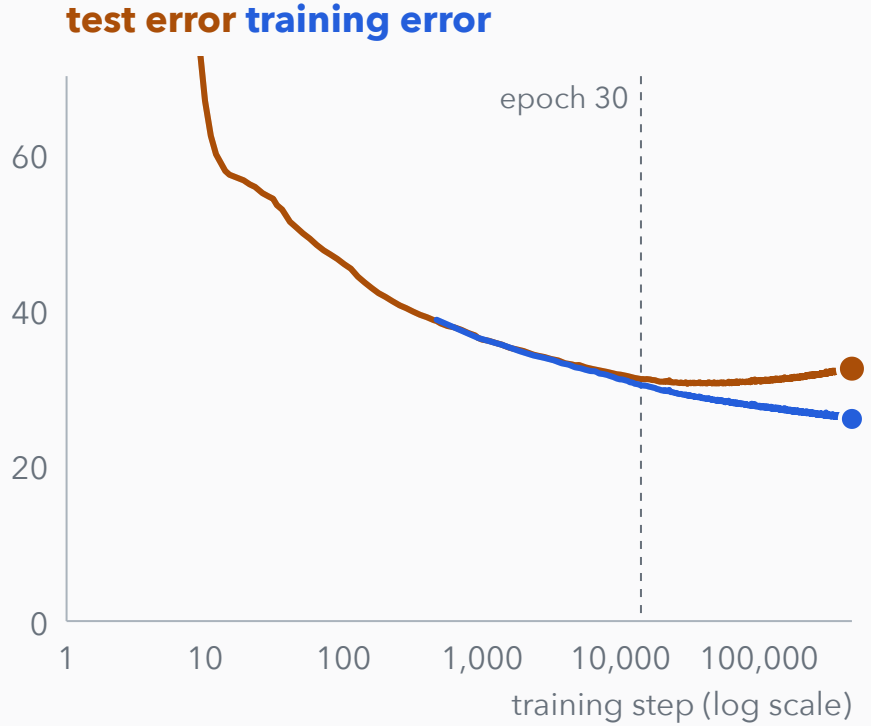


28 3 56 52 21 69 75 15

squared error of each digit (summed over its 784 pixels)

average over the 10,000 test digits: 32.4

average over the 60,000 training digits: 26.0



> Each digit takes shape: 31.1 after 30 epochs, where Lecture 2 stopped. Carry on to 1,000: the training error falls to 26.0, but the test error is lowest at epoch 65 (30.5) and ends at 32.4. Two numbers per digit, not training time, is the limit.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 2.4

More numbers in the code, sharper rebuilds

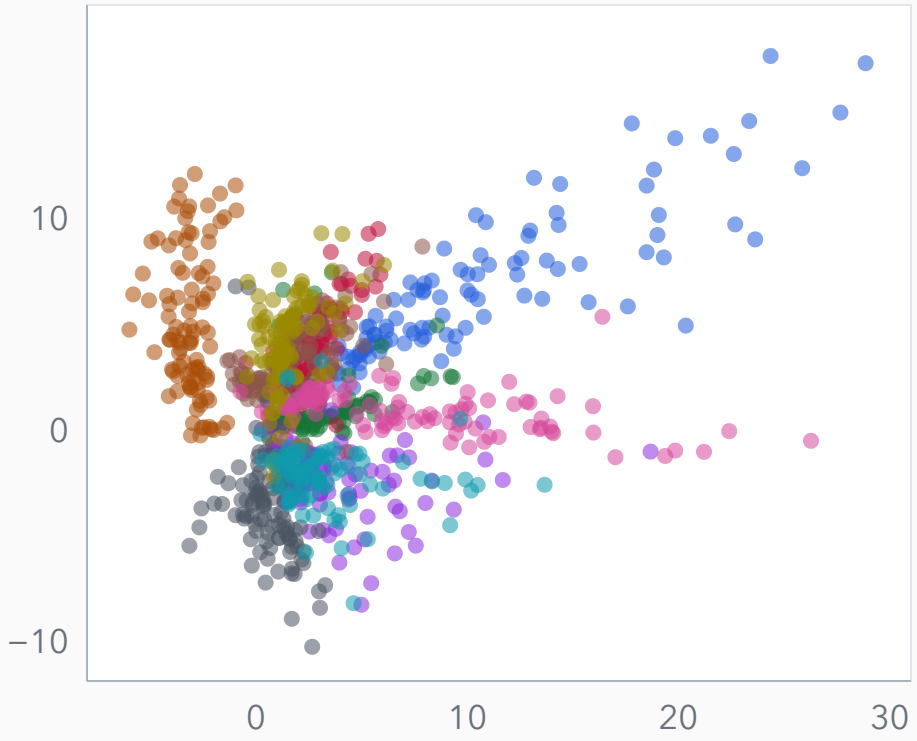


> With 32 and 128 numbers the strokes come back almost exactly. Hold on to this: Part 3 asks which of them can generate.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 2.4

Where the codes land: 1,000 test digits at their 2 numbers



0 1 2 3 4 5 6 7 8 9

the whole code: z_1 and z_2

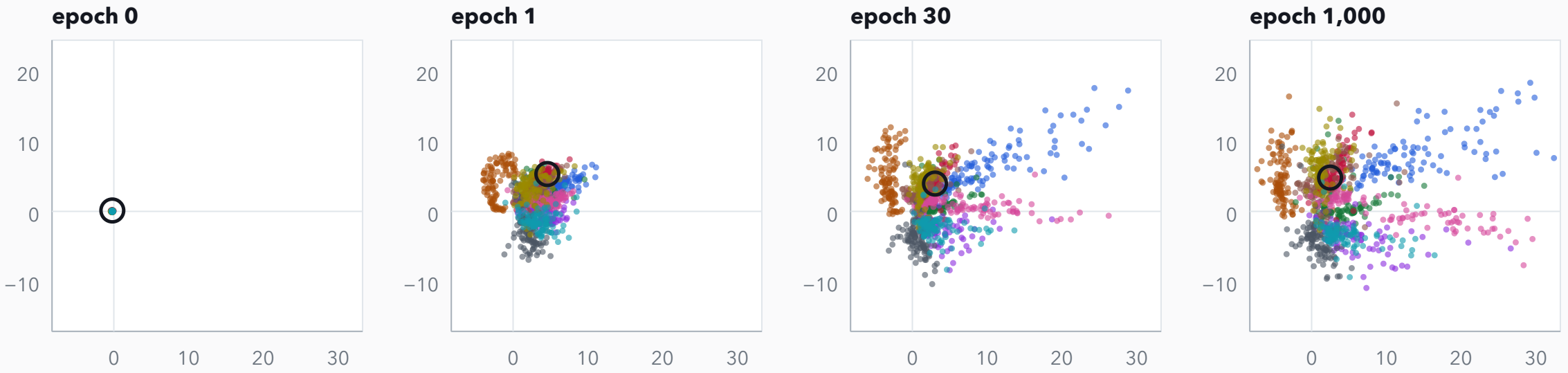
This autoencoder has exactly 2 numbers in its code, so nothing is hidden here. Hover any point to see its digit.

> Digits of one kind cluster together, although no labels were used in training. The codes run from about −5 to 22.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 2.4

During training: clusters form, then the codes drift out



train 184.5 · test 184.5

train 38.7 · test 38.5

train 30.3 · test 31.1

train 26.0 · test 32.4

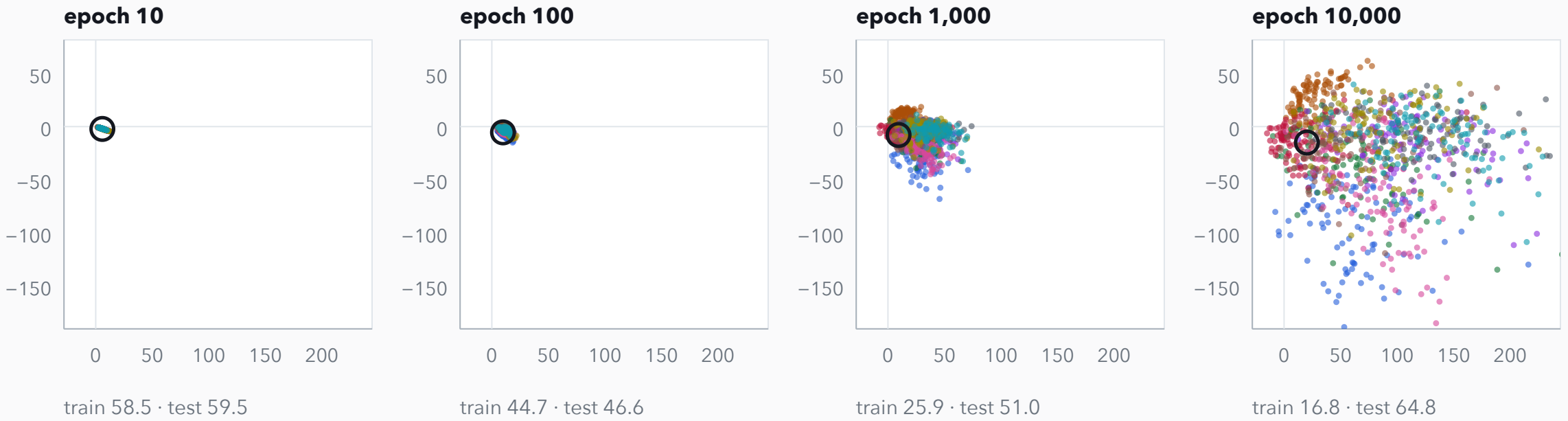
Trained on all 60,000 digits: the same axes in every panel. Our 3 is ringed.

> Press play. Epoch 30 is the last slide's picture; training goes on to 1,000. The clusters sharpen a little (71%) while the cloud keeps stretching: 13 wide after one epoch, 26 at 30, 33 at 300, 35 at 1,000. The training error keeps falling (26.0); the test error is lowest at epoch 65 and creeps up to 32.4.

Lecture 2's network and training, replayed and extended by `lecture5/scripts/epochs.py`.

▶ Notebook 2.4

Overfitting: too few digits, and the codes fly apart



train 58.5 · test 59.5

train 44.7 · test 46.6

train 25.9 · test 51.0

train 16.8 · test 64.8

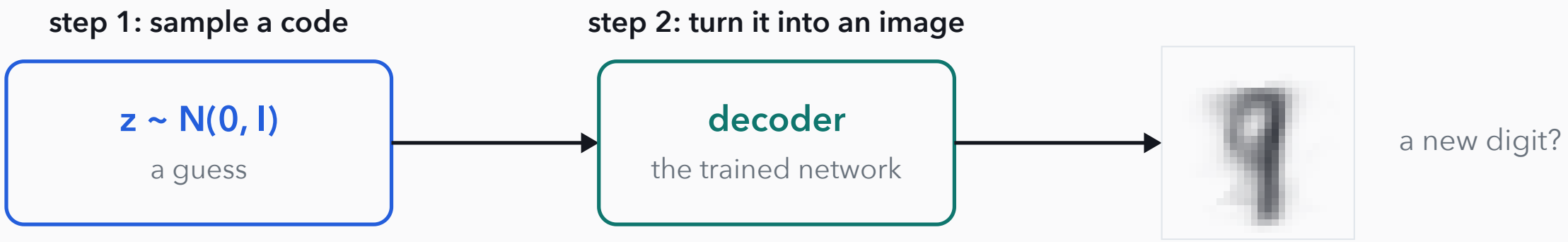
Trained on only 200 digits (20 of each): the same axes in every panel. Our 3 is ringed.

> Press play with “same axes throughout”: the codes fly outward, and the cloud grows from 14 wide at epoch 100 to 231. With so few digits the clusters never formed well (35% same-digit neighbours, against 70% before), and at the end our 3 comes back looking like one of the 200 training 2s.

`lecture5/scripts/epochs.py` (MNIST, fixed seeds).

▶ Notebook

Part 3 · Can an autoencoder generate?



3.1 Decode random codes ▶

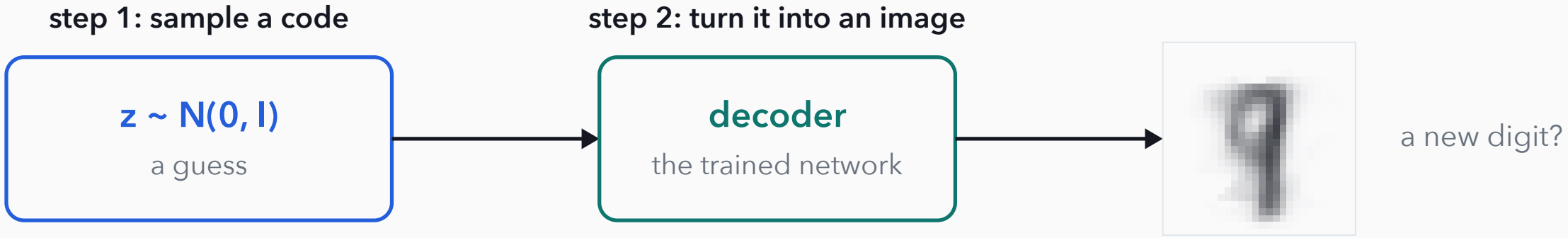
3.2 Where the codes are ▶

3.3 What we want instead ▶

> Keep the decoder, throw the encoder away, and invent the code. Does a random code give a digit?

▶ Notebook 3.1

Keep the decoder, feed it random numbers

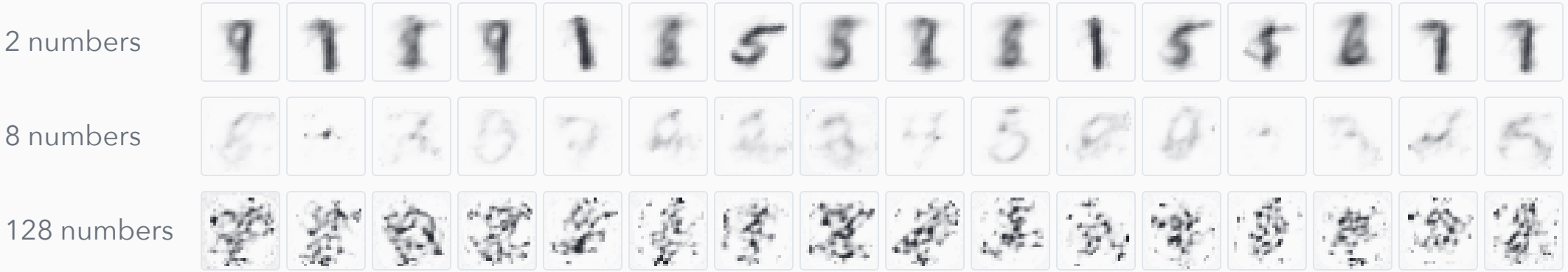


Which random numbers? The simplest: z ~ N(0, I), two independent standard bells. In code: `torch.randn(2)`.

> We need a distribution to draw codes from. Start with the simplest, the standard normal N(0, I).

▶ Notebook 3.1

Random codes through three autoencoders



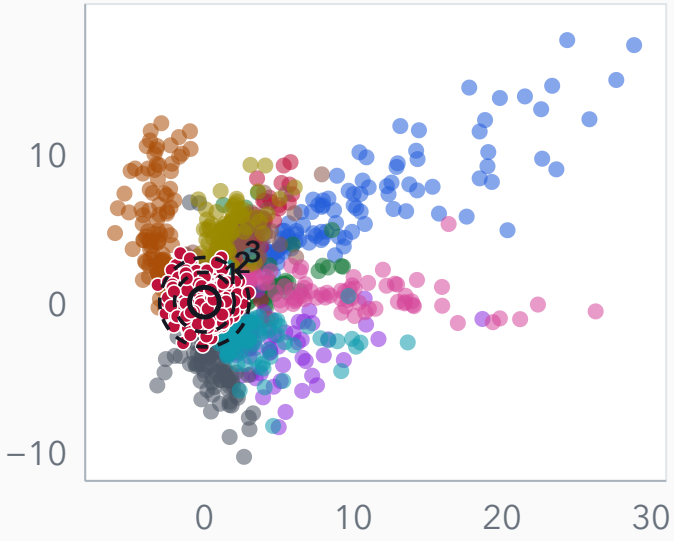
Takeaway. Better reconstruction did not give better samples: the best rebuilder makes static. Why? Look at where the codes are.

128 numbers, the best rebuilder of all, gives static.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 3.2

Put the random codes on top of the real ones



0 1 2 3 4 5 6 7 8 9

real codes and 500 random ones (red)

Circles: N(0, I) at radius 1, 2 and 3.
Hover a dot to see the digit and what the decoder draws from that code.

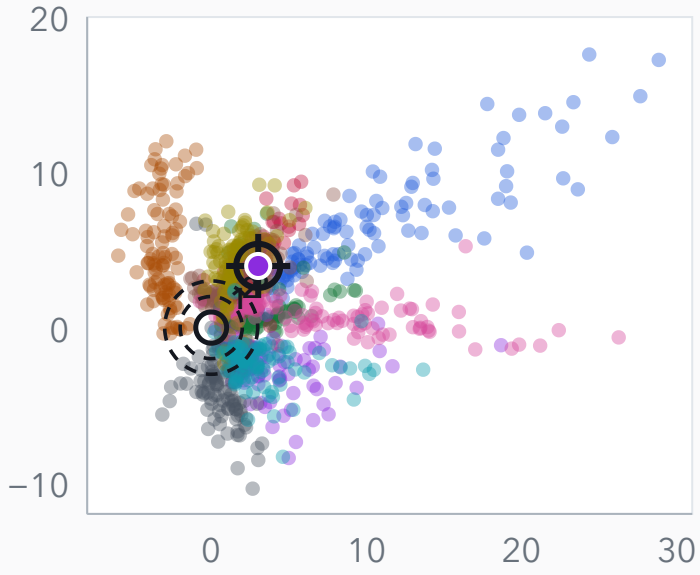
Takeaway. N(0, I) covers a small corner of the real codes: random codes reach a few digit types and miss the rest.

Nothing in the loss asked the codes to sit near N(0, I). The red draws land in one corner: some digits there, none of the rest.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 3.2

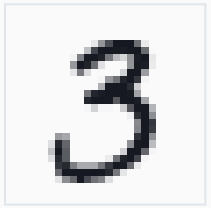
Your turn: three ways to choose a code



0 1 2 3 4 5 6 7 8 9

decoder of the autoencoder, run in your browser





circles: N(0, I)
at radius 1, 2, 3

the digit we encoded

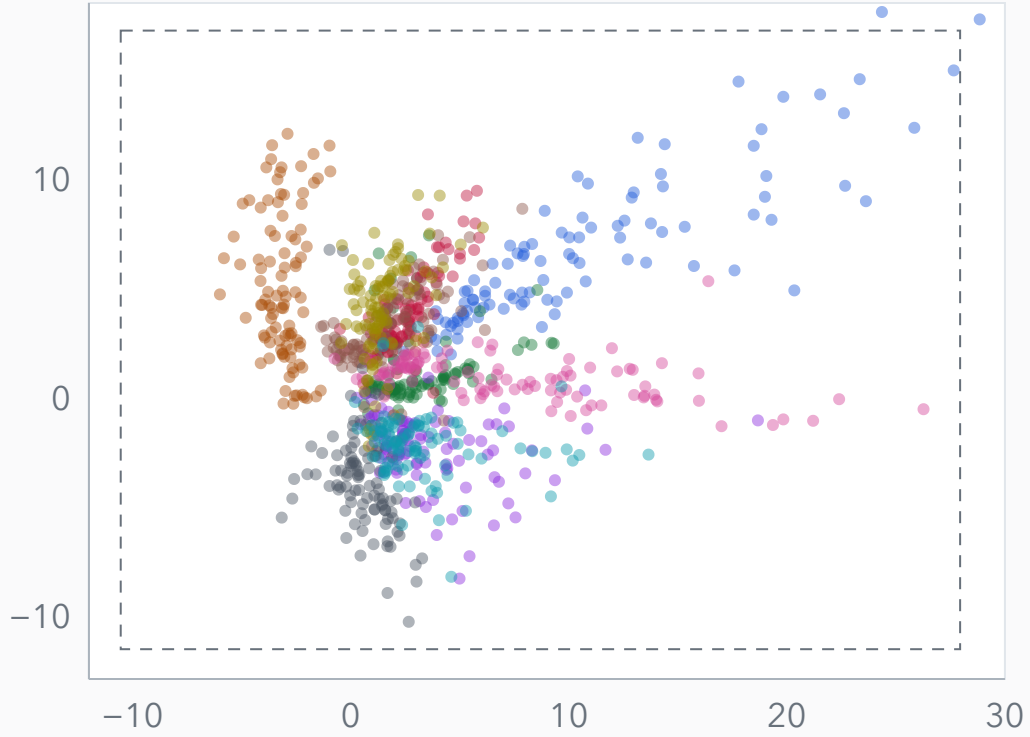
z = (3.02, 3.97)
the code of our 3, decoded

> Draw a random code from N(0, I), encode a real test digit, or click anywhere on the plane. The decoder runs live in the browser.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 3.2

Decode a whole grid of codes





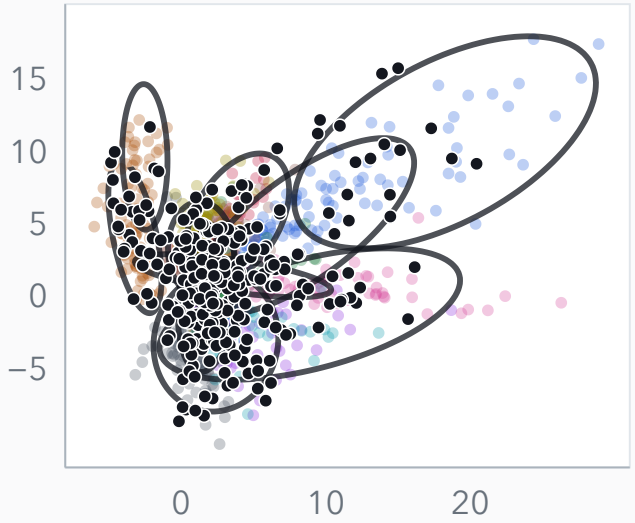
Takeaway. Good digits only where real codes are. Away from them the decoder only extrapolates.

Over the real codes, digits blend into each other. Far from them, thick 1s and 0s and broken strokes: the decoder never trained there.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 3.2

Why not sample from the real codes instead of N(0, I)?



1 · Copy: pick a training digit's code



2 · Jitter: a real code plus N(0, I) noise



3 · Fit 10 Gaussians to the codes, sample them



Takeaway. It works, and large systems do it: Stable Diffusion trains an autoencoder, then a second generative model for its codes. The VAE does it in one model: training puts the codes in N(0, I), and it can score p(x).

So it works, and large systems do it. The VAE gets there in one model and one objective, and also gives a score for p(x).

Lecture 2's 2-number autoencoder and its 10-Gaussian fit to the training codes; draws and decodes: `lecture5/scripts/empirical.py`.

▶ Notebook 3.3

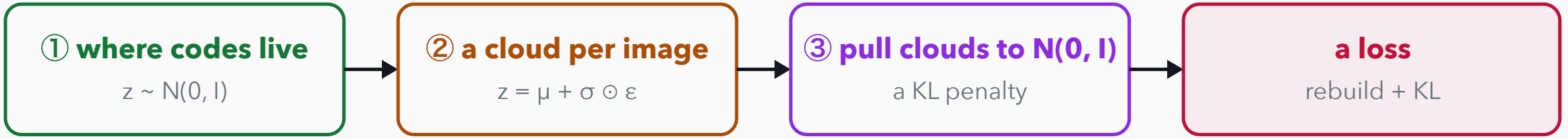
What we saw, and what we want instead

what we saw	what we want
random N(0, I) codes miss the real codes	codes that fill a region we know how to sample
gaps between codes decode to junk	a decoder trained everywhere in that region, gaps included
no way to say how likely an image is	one probability model that can score and sample

> Problem three: the autoencoder cannot score an image. We want a probability model. Part 4 fixes the first two; Part 5 shows the fix is that model.

▶ Notebook

Part 4 · From an autoencoder to a VAE: three fixes



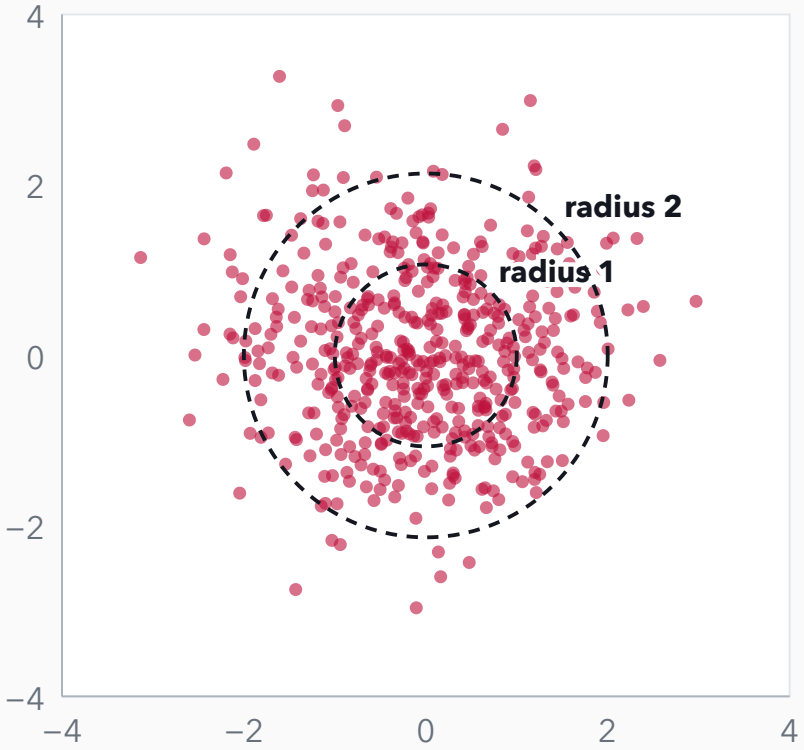
Part 5: this loss is exactly minus Lecture 1's ELBO, one image at a time

4.1 Fix 1: choose where codes live ▶
4.2 Fix 2: a cloud for each image ▶
4.3 Fix 3: pull the clouds to N(0, I) ▶
4.4 The loss the fixes give ▶

> The autoencoder cannot generate. Three fixes, one at a time, turn it into a VAE; each one answers a problem we saw in Part 3.

▶ Notebook 4.1

Fix 1: choose where the codes live, before training



Decide the distribution of codes in advance: z ~ N(0, I), two independent standard bells.

Why this one? It is easy to sample (`torch.randn(2)`), it has no gaps, and it is the same in every training run.

Generating becomes two lines: draw z, decode it. The work moves into training: the codes of real digits must fill this disc.

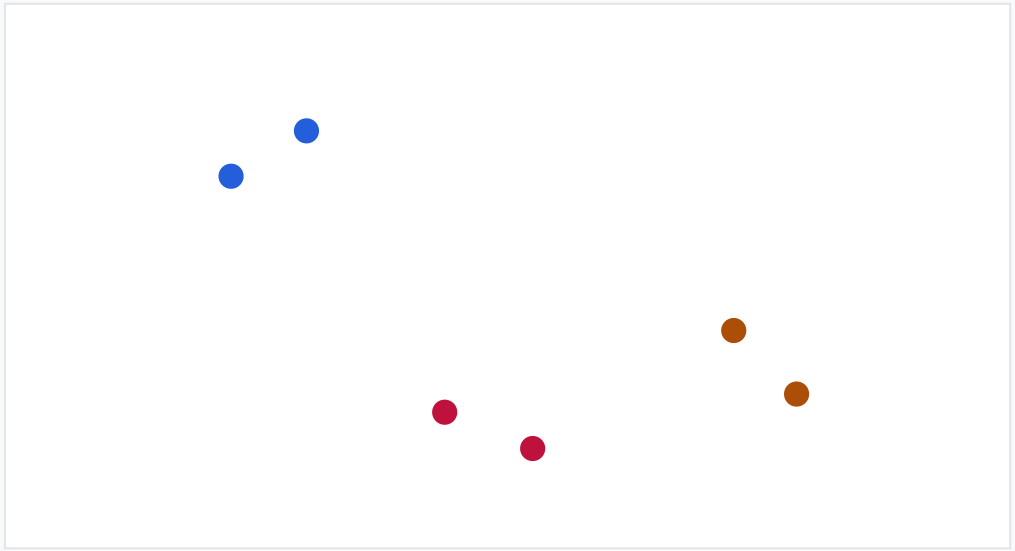
> If training makes the codes of real digits fill this disc, every random draw lands near some digit. Fixes 2 and 3 make that happen.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 4.2

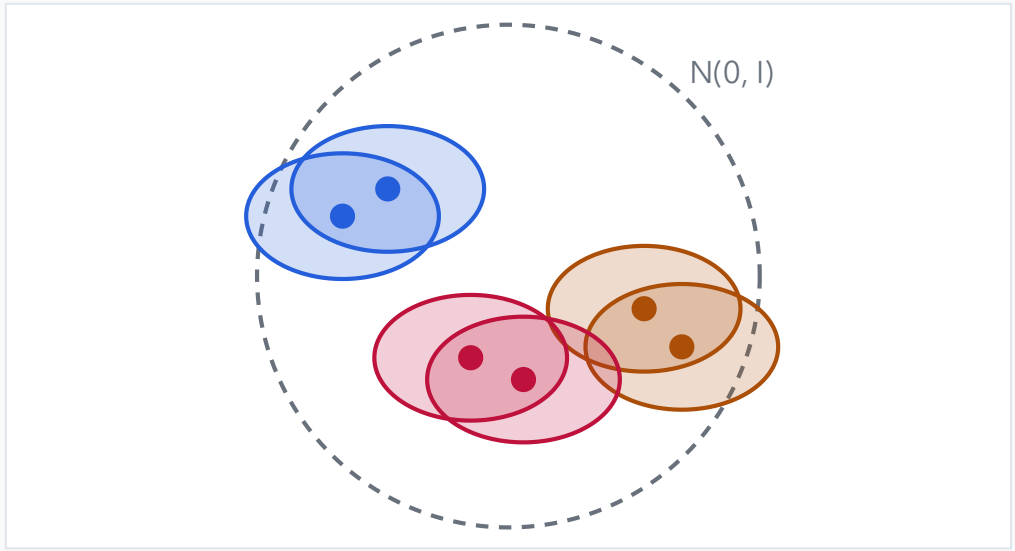
Fix 2: send each image to a small cloud of codes, not a point

autoencoder: each image → one point



points: the space between them is never trained

VAE: each image → a small cloud



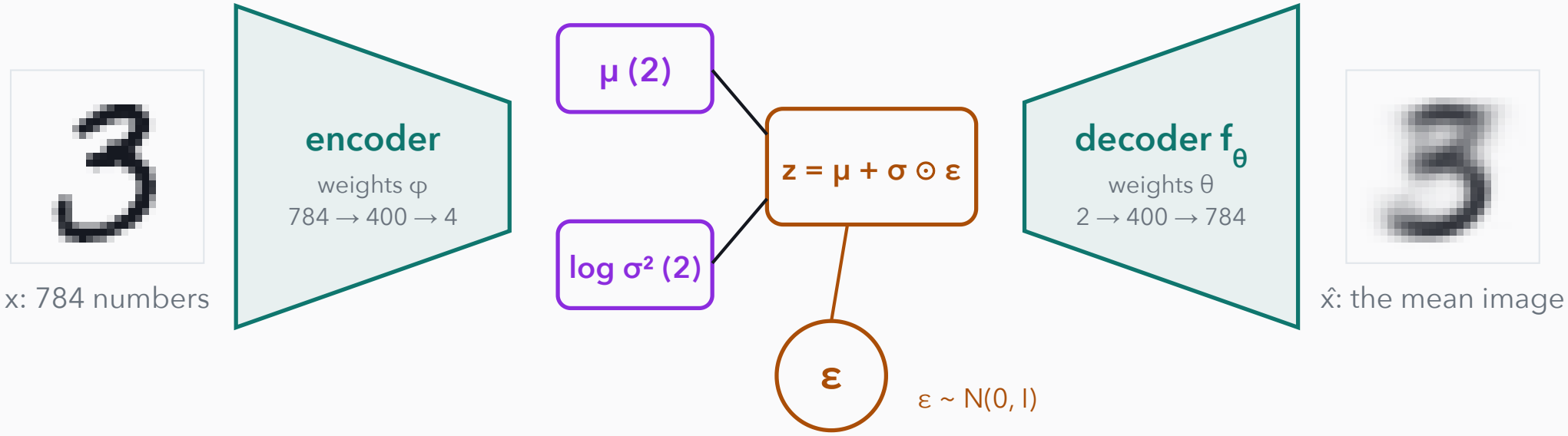
clouds overlap: the space between images is trained

A sketch, not data. Colours: three kinds of digit.

> Instead, send each image to a small cloud and decode a random point from it. Neighbouring clouds overlap, so the decoder learns the gaps too.

▶ Notebook 4.2

How a network outputs a cloud: a mean, a spread and a random draw



The decoder is the autoencoder's, unchanged: from z to a mean image.

▶ Notebook 4.3

Fix 3: a penalty that pulls every cloud towards N(0, I)

With Fixes 1 and 2 alone, nothing stops the encoder from shrinking each cloud to a point and placing it far from the centre: an autoencoder again.

$$\text{penalty} = \mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0,1)\big) = \tfrac{1}{2}\big(\mu^2 + \sigma^2 - \log\sigma^2 - 1\big)$$

↗ Lecture 1, 2.6: tool ②: the Gaussian KL

one code number's cloud	μ	σ	penalty
exactly N(0, 1)	0	1	0
far from the centre	5	1	12.5
shrunk almost to a point	0	0.01	4.1
our 3, first code number	−0.21	0.013	3.9

> Far away costs (μ = 5: 12.5); a cloud shrunk to a point costs too (σ = 0.01: 4.1). Only N(0, 1) itself is free.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 4.4

The three fixes give a loss

$$L = \underbrace{\left\| x - f_\theta(\mu + \sigma \odot \epsilon) \right\|^2}_{\text{rebuild from a draw of the cloud (Fix 2)}} + \underbrace{\mathrm{KL}\left(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, I)\right)}_{\text{pull the cloud to } \mathcal{N}(0,I) \text{ (Fixes 1 and 3)}}$$

Two questions remain. How much weight should each term get? And is this a sensible objective, or a hack that happens to work?

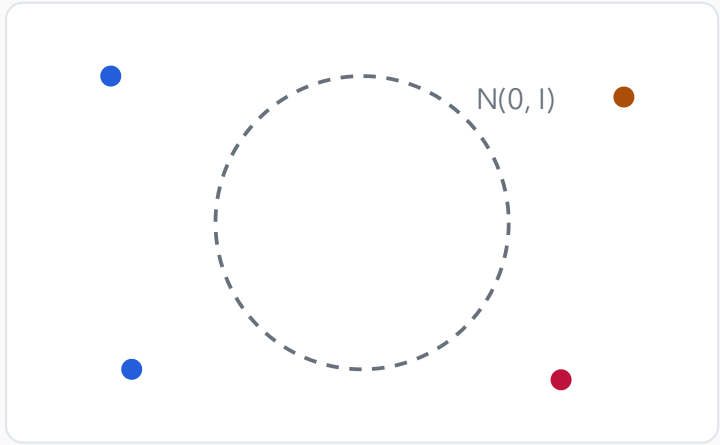
Part 5 answers both with Lecture 1: this sum is minus the ELBO of a probability model, with a weight of 1/2σ² on the first term.

> Lecture 1 answers both: the sum is minus the ELBO, the weight comes from the pixel noise, and the KL appears by itself.

▶ Notebook 4.4

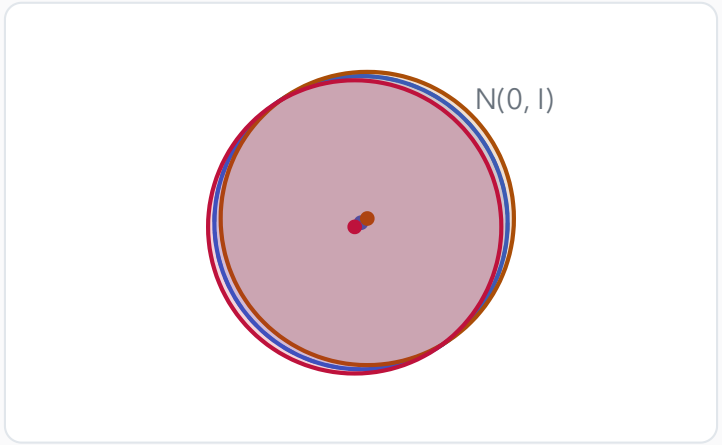
Two terms pull in opposite directions

reconstruction alone



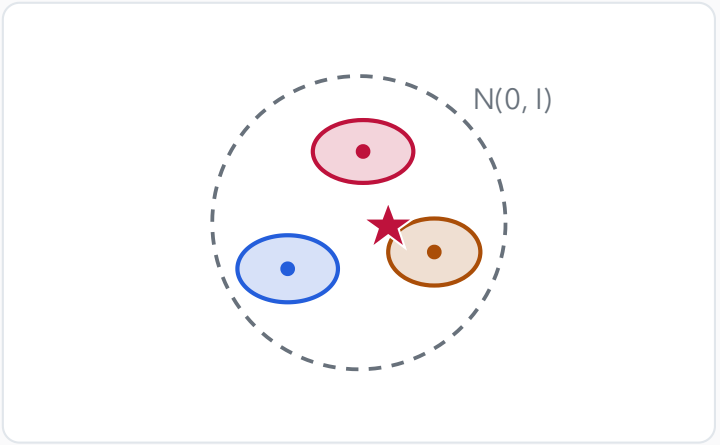
tiny clouds, far apart:
an autoencoder again

KL alone



every image gets the same cloud N(0, I):
z says nothing about the image

both together



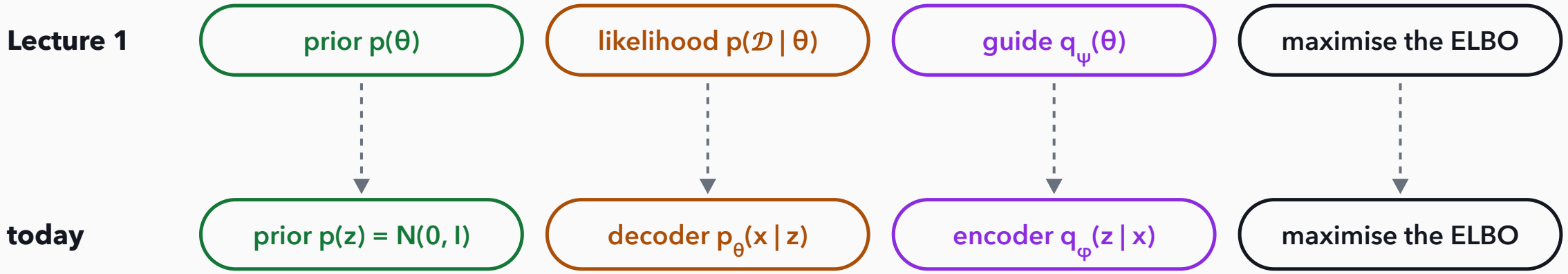
clouds near the centre, still apart:
a random z (star) lands near some cloud

Takeaway. A VAE is an autoencoder whose codes are noisy clouds held near N(0, I). To generate, draw z ~ N(0, I) and decode.

So a random draw from N(0, I) usually lands on or near some cloud, and the decoder has been trained to draw something there.

▶ Notebook

Part 5 · Why this loss: Lecture 1's ELBO, one image at a time



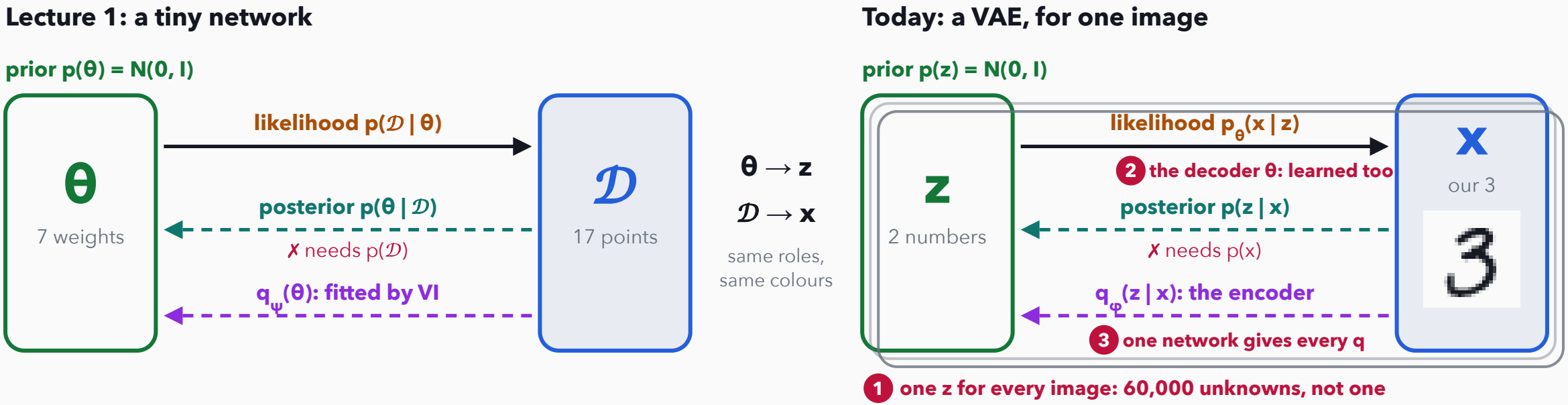
swap the names: the weights θ become a code z for each image, and the data $\mathcal{D}$ become one image x

5.1 Same problem as Lecture 1 ▶
5.2 A model with a hidden code ▶
5.3 The same wall as Lecture 1 ▶
5.4 The variational approximation ▶
5.5 The loss, derived from the ELBO ▶
5.6 Both terms, live ▶
5.7 What the VAE learned ▶
5.8 The KL weight β ▶

> Why is Part 4's loss the right one? Write a model with a hidden code, hit Lecture 1's wall, and get past it the same way: the ELBO, one image at a time.

▶ Notebook 5.1

Same problem as Lecture 1: the same picture, new names



$$\log p(\mathcal{D}) = \mathrm{ELBO}(q) + \mathrm{KL}\big(q \,\|\, p(\theta \mid \mathcal{D})\big)$$

$$\log p_\theta(x) = \mathrm{ELBO}(x) + \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z \mid x)\big)$$

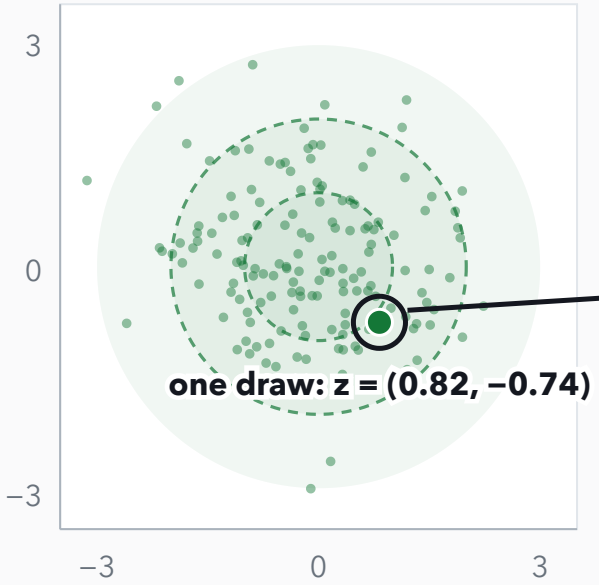
> Three differences: ① every image has its own z, 60,000 unknowns; ② the likelihood has weights of its own, the decoder, learned at the same time; ③ one network, the encoder, gives every image its q.

▶ Notebook 5.2

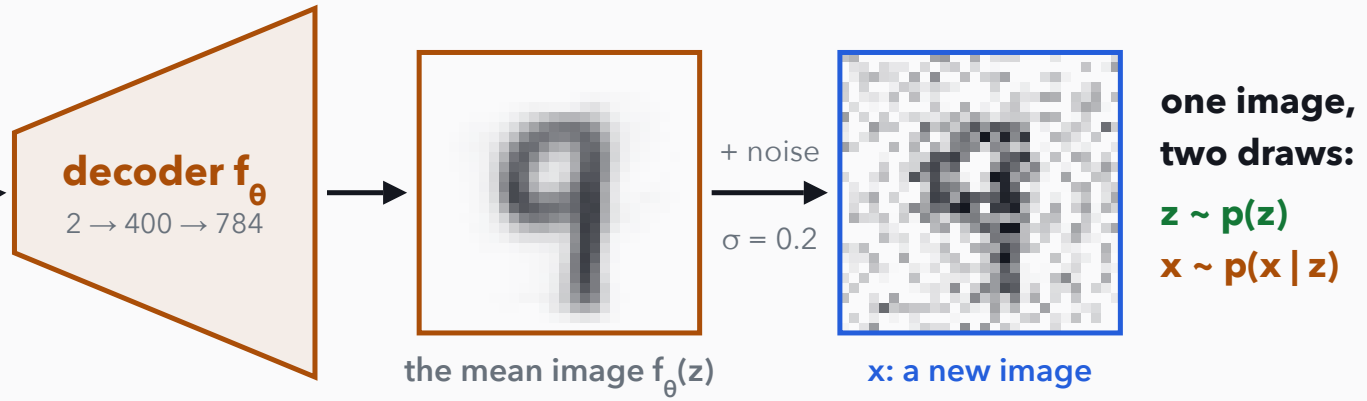
The model's story: first a code from the prior, then an image

① the prior p(z) = N(0, I)

where codes come from, before any image



② the likelihood p(x | z): decode z, then add pixel noise



one image, two draws:

z ~ p(z)

x ~ p(x | z)

$$p_\theta(x, z) \;=\; p(z) \;\cdot\; p_\theta(x \mid z)$$

> Then every pixel gets its own noise of size σ = 0.2. Two draws make one image: z from the prior, then x from the likelihood. Their product is the joint p(x, z).

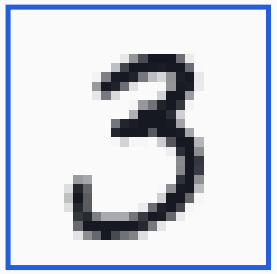
Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.2

The likelihood scores a code: how well does it draw our 3?

$$\log p_\theta(x \mid z) = \sum_{j=1}^{784} \log \mathcal{N}\left(x_j;\ f_\theta(z)_j,\ \sigma^2\right) = -\frac{\|x - f_\theta(z)\|^2}{2\sigma^2} + 541.4$$

the image



x: our 3

in the posterior's patch



z = (−0.23, 0.06)

squared error
23.1

log p(x | z)
252

a draw from the prior

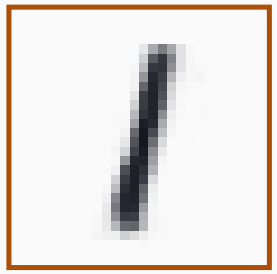


z = (0.82, −0.74)

squared error
72.1

log p(x | z)
−360

where the 1s are



z = (0.00, 3.20)

squared error
83.7

log p(x | z)
−505

the decoder's mean image $f_\theta(z)$ for each code: a code that draws our 3 well gets a high likelihood

σ = 0.2: about the typical pixel error of a 2-number code, √(31 / 784) ≈ 0.2. The constant is −784 log(σ√2π).

> σ = 0.2 is a modelling choice, matched to a typical pixel error. Smaller σ makes every unit of squared error cost more.

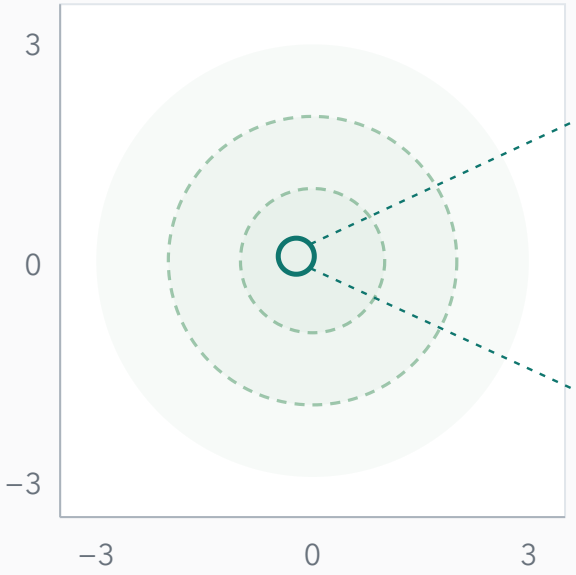
Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.3

Run the story backwards: which codes made our 3?

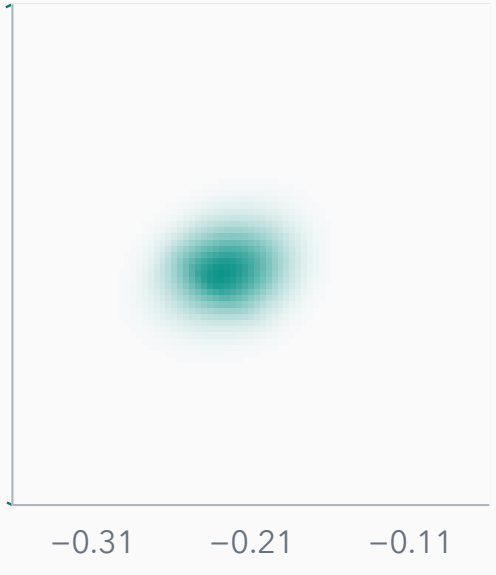
the code plane

green: the prior N(0, I)



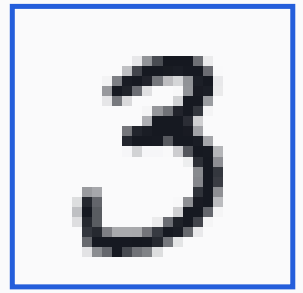
the posterior p(z | x), zoomed in 23×

shade: how well each code explains our 3



Bayes' rule

given an image



x: our 3, observed

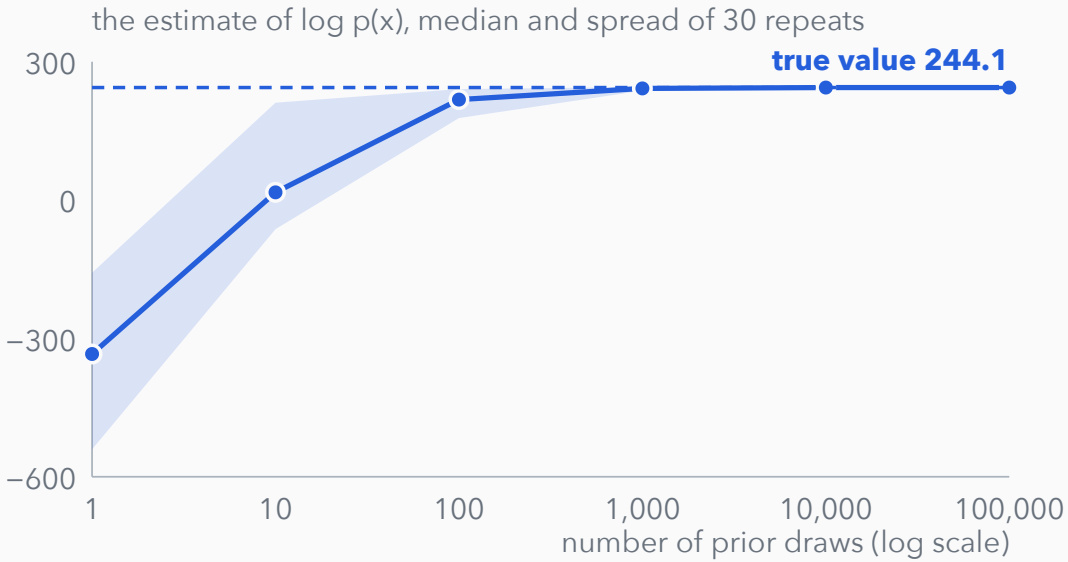
$$p(z \mid x) = \frac{p_\theta(x \mid z)\, p(z)}{p_\theta(x)} \qquad p_\theta(x) = \int p_\theta(x \mid z)\, p(z)\, dz$$

> The denominator, the evidence p(x), adds up the likelihood of every code, weighted by its prior: an integral through a neural network, with no formula. Lecture 1's wall exactly.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.3

The evidence p(x): average the likelihood over the prior



$$p_\theta(x) = \mathbb{E}_{z\sim p(z)}\left[p_\theta(x \mid z)\right] \approx \frac{1}{S}\sum_{s=1}^{S} p_\theta(x \mid z_s)$$

Draw S codes from the prior, decode each one, and average the likelihoods.

Of 100,000 prior draws, the best 10 carry 38% of the total. Almost every code from the prior explains our 3 terribly.

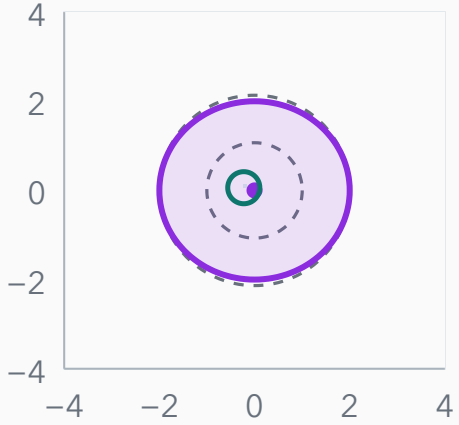
> Why so many? The codes that explain our 3 form a patch about 0.018 wide, and random codes almost never land in it. For one image in 2 numbers brute force barely works; for 60,000 images, every step, in 16 numbers, it is hopeless.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.4

The variational approximation: the closest Gaussian

q = N(0, I), the prior

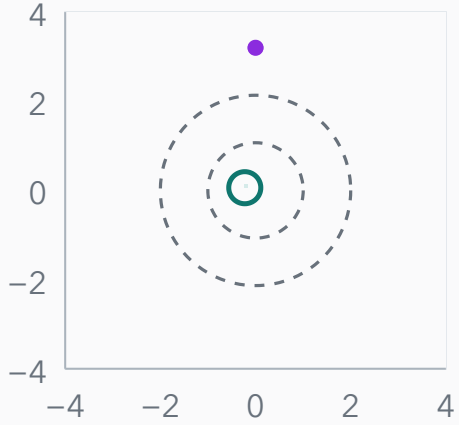


ELBO −269.7

gap 514

teal ring: the posterior

q on the 1s

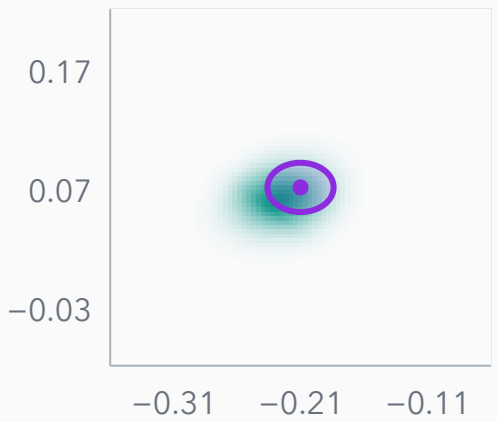


ELBO −514.2

gap 758

teal ring: the posterior

the encoder's q

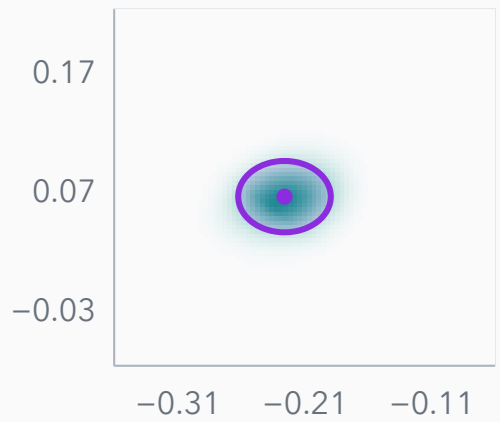


ELBO 243.4

gap 0.7

teal: the posterior

q fitted by VI (5.4)



ELBO 244.0

gap < 0.1

teal: the posterior

$$\log p_\theta(x) = \mathrm{ELBO}(q) + \mathrm{KL}\big(q \,\|\, p(z \mid x)\big)$$

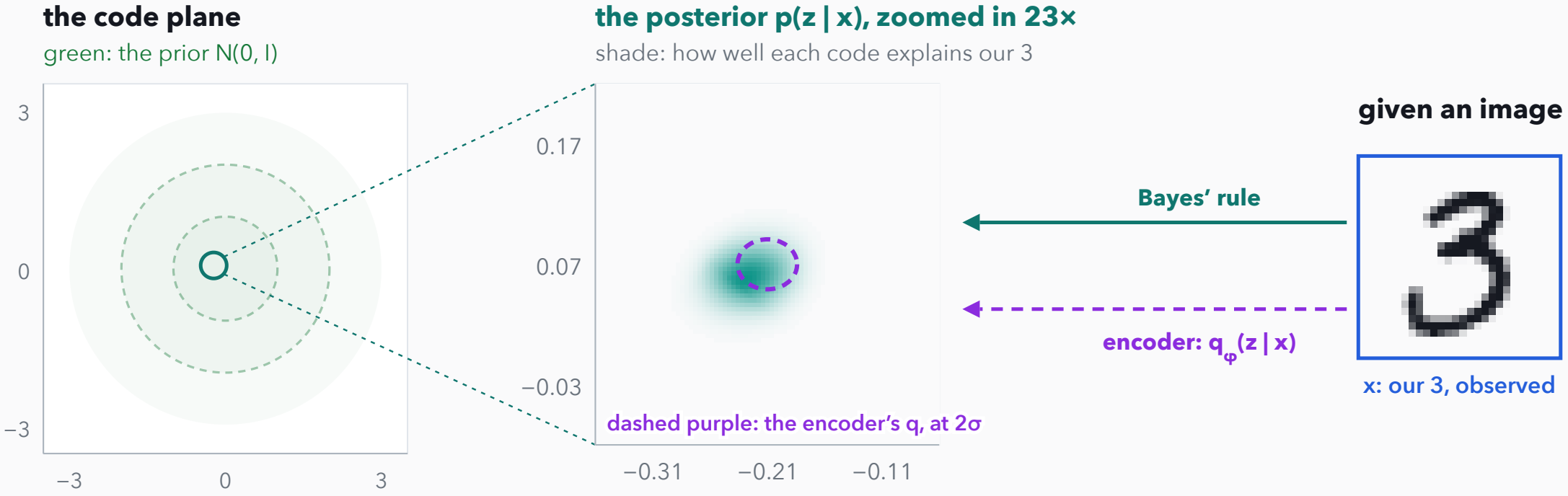
↗ Lecture 1, 1.5: approximate the posterior

> How do we know which is closest, without the posterior? log p(x) is one fixed number, so the gap log p(x) − ELBO is exactly the KL to the posterior. Highest ELBO = closest q: gaps 514, 758, 0.7 and < 0.1.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.4

The encoder's cloud is q: one network gives q for every image



Generative model: z → x, the decoder. Variational approximation: x → z, the encoder. One objective trains both: the ELBO.

> Decoder θ and encoder φ are trained together, by one objective: the ELBO, summed over all images.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.5

From the ELBO to the loss: recon + KL, derived, not guessed

start
Lecture 1's ELBO, names swapped

choice 1
pixels: Gaussian around the decoder's output, noise 0.2

choice 2
prior N(0, I); q from the encoder: N(μ, σ²)

result
minus the ELBO: Part 4's loss

$$\mathrm{ELBO}(x) = \mathbb{E}_{z\sim q_\phi(z|x)}\big[\log p_\theta(x \mid z)\big] - \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z)\big)$$

$$\log p_\theta(x \mid z) = -\frac{\|x - f_\theta(z)\|^2}{2 \cdot 0.2^2} + 541.4$$

$$\mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, I)\big) = \tfrac{1}{2}\sum_k \left(\mu_k^2 + \sigma_k^2 - \log \sigma_k^2 - 1\right)$$

$$-\mathrm{ELBO}(x) = \underbrace{\mathbb{E}_q\left[\tfrac{1}{2\cdot 0.2^2}\|x - f_\theta(z)\|^2\right]}_{\text{rebuild from the cloud: Fix 2}} + \underbrace{\mathrm{KL}\big(q_\phi \,\|\, \mathcal{N}(0, I)\big)}_{\text{pull to } \mathcal{N}(0,I)\text{: Fixes 1, 3}} - 541.4$$

> Flip the sign. Minus the ELBO is the rebuild error plus the KL: Part 4's loss, term for term, now with its weight decided. Nothing was guessed; every term comes from a modelling choice.

▶ Notebook 5.5

One step for our 3, with numbers

$$\mu = (-0.211,\ 0.070), \quad \sigma = (0.013,\ 0.010), \quad \epsilon = (0.5,\ -1.0)$$

$$z = \mu + \sigma \odot \epsilon = (-0.204,\ 0.060)$$



vs



$$\frac{1}{2(0.2)^2} \times 23.18 = 289.8$$

$$\mathrm{KL} = 7.94 \qquad L = 289.8 + 7.94 = 297.7$$

Add the KL to the prior, from tool ②: the loss for this image on this step.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.5

The VAE in PyTorch: three changes to the autoencoder

```
class VAE(AutoEncoder):                # same decoder
    def __init__(self, z=2):
        super().__init__(z)
        self.encoder[-1] = nn.Linear(400, 2 * z)  # (1)

    def forward(self, x):
        mu, logvar = self.encoder(x).chunk(2, dim=1)
        eps = torch.randn_like(mu)
        z = mu + torch.exp(0.5 * logvar) * eps       # (2)
        return self.decoder(z), mu, logvar

x_hat, mu, logvar = model(x)
recon = ((x_hat - x)**2).sum(1) / (2 * 0.2**2)
kl = 0.5 * (mu**2 + logvar.exp() - logvar - 1).sum(1)  # (3)
loss = (recon + kl).mean()
```

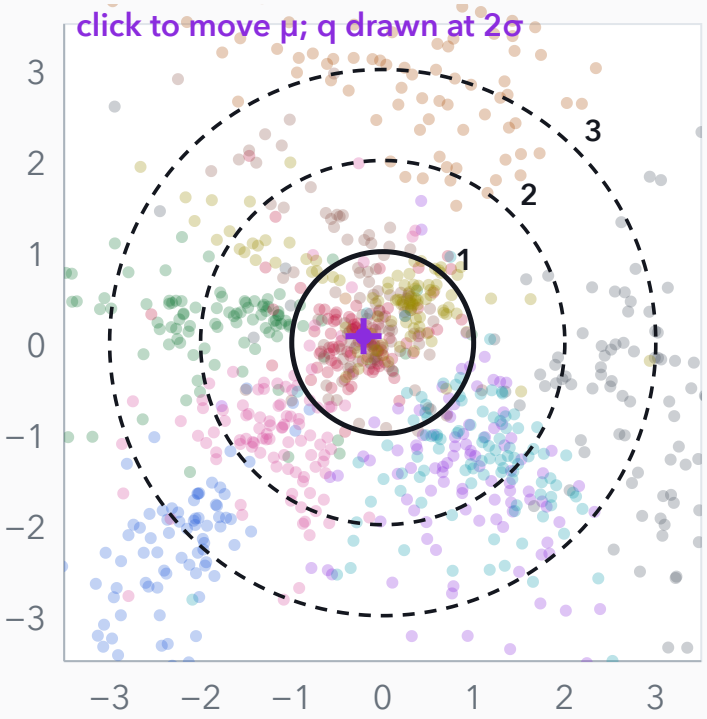
(1) the encoder outputs 2z numbers, (2) the code is a reparameterised draw (tool ①), (3) the loss adds the KL (tool ②).

The training loop is the autoencoder's, unchanged.

> Everything else, the batches, Adam and the passes over the data, is the autoencoder's loop.

▶ Notebook 5.6

Your turn: choose q for our 3, and watch both terms



x: our 3 **f(μ): decoded** five draws z from q, decoded



reconstruction	**290.1**	E‖x − f(z)‖² / 2σ²: small if draws rebuild our 3
KL to the prior	**7.9**	KL(q ‖ N(0, I)): small if q sits near N(0, I)
ELBO	**243.3**	= 541.4 − reconstruction − KL

gap to log p(x): 0.7, the KL from q to the posterior

log p(x) = 244.1: the ceiling

purple dot: the ELBO · red: the gap

−600 −300 0 300

q = the encoder's q for our 3: μ = (−0.21, 0.07), σ = 0.012.

> Try q = N(0, I), then q on the 1s. Every q stays below the ceiling log p(x); the red gap is the KL to the posterior.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.7

Now sample: z ~ N(0, I), then decode

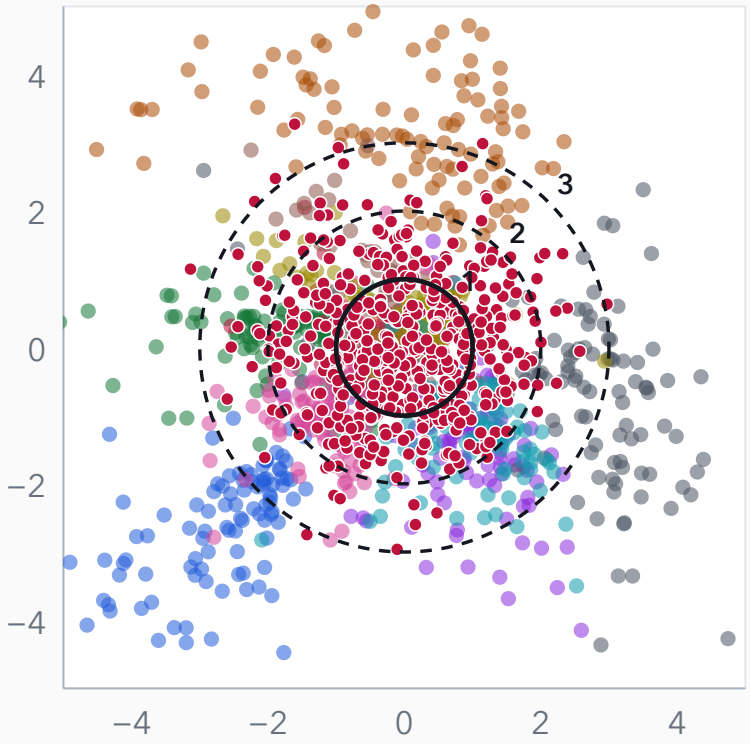


> Through the 16-number VAE: sharper and more varied. The fakes at the start of the hour came from this model.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.7

The codes now fill N(0, I)



Codes of all 10,000 test digits: mean (−0.2, −0.2), spread (1.8, 1.7).

Random draws (red) now land among real codes, on every digit.

Overlay 500 draws from N(0, I): they fall on real codes of every digit. That is why sampling works.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.7

Decode a grid of codes: digits everywhere

VAE: codes from −3 to 3 in each direction



walk in a straight line from a 1 to a 0



the same walk, 16-number VAE

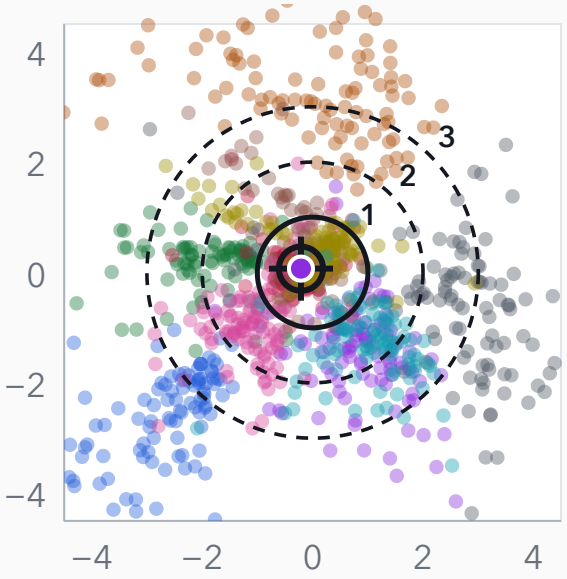


> Walk from a 1 to a 0: every step is a plausible digit. The space between images was trained, because each image owned a cloud.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.7

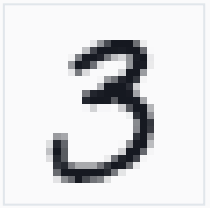
Your turn again: choose a code for the VAE



0 1 2 3 4 5 6 7 8 9

decoder of the VAE, run in your browser





the digit we encoded

circles: N(0, I)
at radius 1, 2, 3

z = (−0.21, 0.07)
the code of our 3, decoded

> Random codes, encoded test digits, or click anywhere. Switch between the VAE and the autoencoder to compare.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.8

Put a weight on the KL: β

$$L_\beta = \mathbb{E}_q\left[\frac{1}{2\sigma^2}\left\|x - f_\theta(z)\right\|^2\right] + \beta\,\mathrm{KL}\left(q_\phi(z \mid x) \,\|\, \mathcal{N}(0, I)\right)$$

β = 1 is the true ELBO. Changing β is the same as assuming noisier or cleaner pixels, a different σ.

Small β

codes free to spread: close to an autoencoder

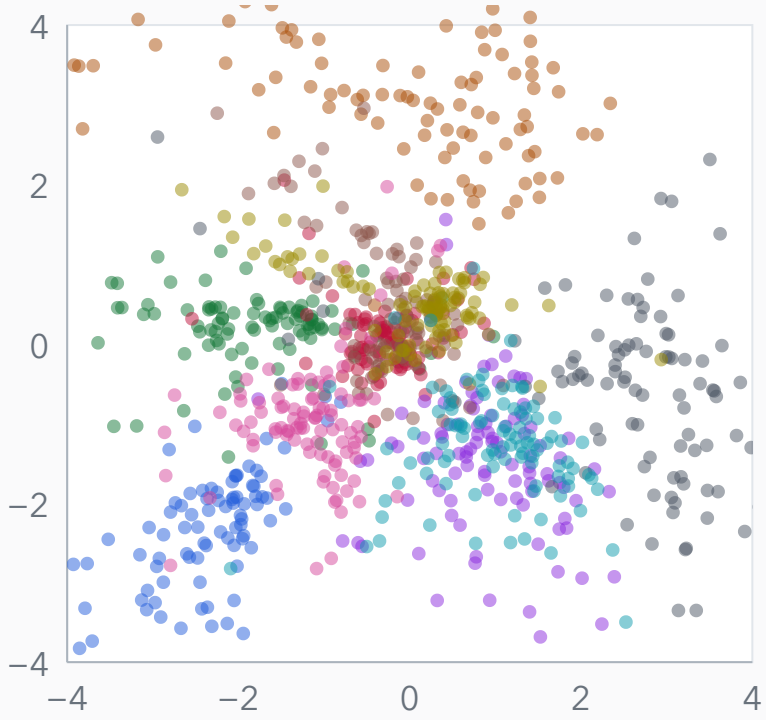
Large β

codes squeezed towards the prior: closer to N(0, I), less detail

> Predict before we look: what happens at β = 0, and at very large β?

▶ Notebook 5.8

Seven VAEs, β from 0 to 256: try them

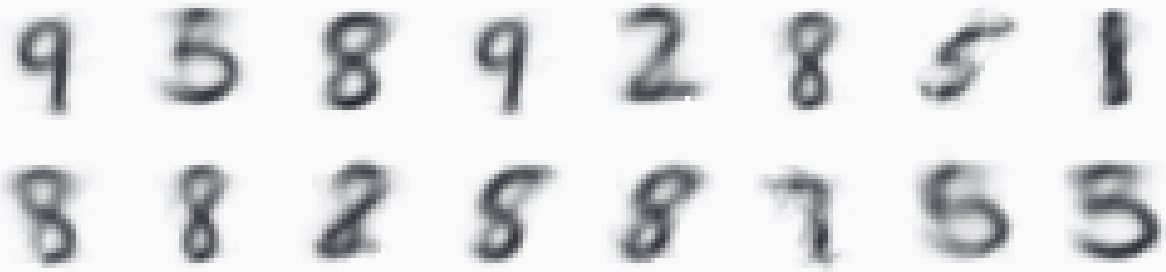


β = 1 KL 9.0 · squared error 30.8 · average σ of q 0.034

reconstructions of 8 test digits



16 samples: z ~ N(0, I), then decode



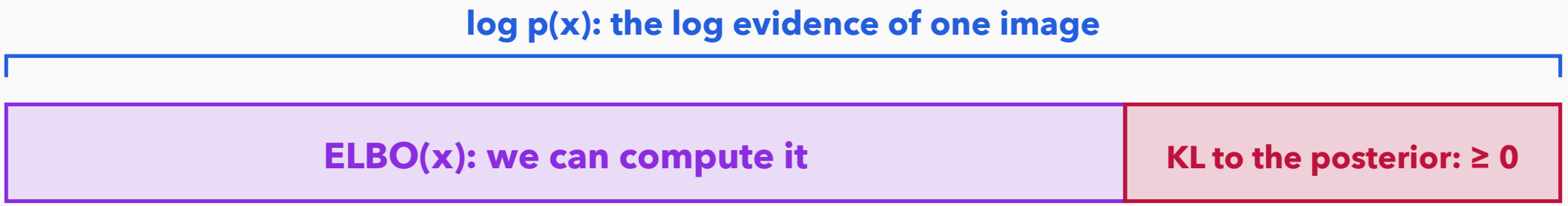
distance between two 0s ÷ distance from a 0 to a 1: 0.23

Click through β. Watch how spread the codes are, the reconstructions, and the samples.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

Part 6 · Putting it together



raise the ELBO, and the KL to the posterior must fall

6.1 The whole lecture in maths ▶

6.2 Wrap-up ▶

> Every piece of the lecture on one page: the VAE's loss, term by term, traced back to KL and the ELBO.

▶ Notebook 6.1

The whole lecture in five lines of maths

		the maths	where we met it
①	KL	$\mathrm{KL}(q \,\|\, p) = \mathbb{E}_q\big[\log q - \log p\big] \;\geq\; 0$	Lecture 1, 2.4
②	the budget	$\log p_\theta(x) = \mathrm{ELBO}(x) + \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z \mid x)\big)$	Lecture 1, 4.1 · today 5.4
③	its two terms	$\mathrm{ELBO}(x) = \mathbb{E}_q\big[\log p_\theta(x \mid z)\big] - \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z)\big)$	Lecture 1, 4.1 · today 5.4
④	Gaussian pixels	$-\log p_\theta(x \mid z) = \frac{1}{2\sigma^2}\|x - f_\theta(z)\|^2 + \text{const}$	today 5.2 and 2.3
⑤	the two tools	$z = \mu + \sigma \odot \epsilon, \qquad \mathrm{KL}\text{ to } \mathcal{N}(0,1) = \frac{1}{2}\big(\mu^2 + \sigma^2 - \log\sigma^2 - 1\big)$	Lecture 1, 3.5, 2.6 · today 4.2–4.3

> ⑤ The two tools make the loss computable: draw z by reparameterisation, and take the KL by formula.

▶ Notebook 6.1

Where each term of the VAE's loss comes from

reconstruction

$$\|x - f(\mu + \sigma \odot \varepsilon)\|^2 / 2\sigma^2$$

- Gaussian pixels: it is $-\log p(x \mid z)$ + const — **5.2**
- the autoencoder's squared error, rescaled — **2.3**
- $z = \mu + \sigma \odot \varepsilon$: Fix 2, tool ① — **Lecture 1, 3.5**

+

KL(q(z | x) ‖ N(0, I))

$$\tfrac{1}{2} \Sigma (\mu^2 + \sigma^2 - \log \sigma^2 - 1)$$

- Fix 3; the formula is tool ② — **Lecture 1, 2.6**
- to the prior p(z) = N(0, I): Fix 1 — **4.1**
- a KL: extra cost, never negative — **Lecture 1, 2.4**

together: −ELBO(x) + const, Lecture 1's objective for one image (5.4)

a smaller loss means a higher lower bound on log p(x), and q closer to the posterior

> Together they are minus Lecture 1's ELBO for one image. Making the loss smaller pushes up a lower bound on log p(x).

▶ Notebook 6.2

Autoencoder and VAE, side by side

	autoencoder	VAE
the encoder outputs	a code z: 2 numbers	a cloud: μ and log σ^2, that is $q_\phi(z \mid x)$
the decoder gets	z	a draw $z = \mu + \sigma \odot \varepsilon$ (tool ①)
the loss	squared error	squared error / $2\sigma^2$ + KL to N(0, I) (tool ②) = −ELBO + const
random z ~ N(0, I) gives	mostly junk	digits
can it score an image?	no	yes, through the ELBO, a lower bound on log p(x)

Takeaway. A VAE is an autoencoder trained with Lecture 1's objective. The KL to the prior makes the codes fill N(0, I), so random codes decode to digits.

Both changes come from Lecture 1: the ELBO with a code per image, the reparameterisation trick, and the Gaussian KL.

▶ Notebook 6.2

Next: can we drop the pixel likelihood altogether?

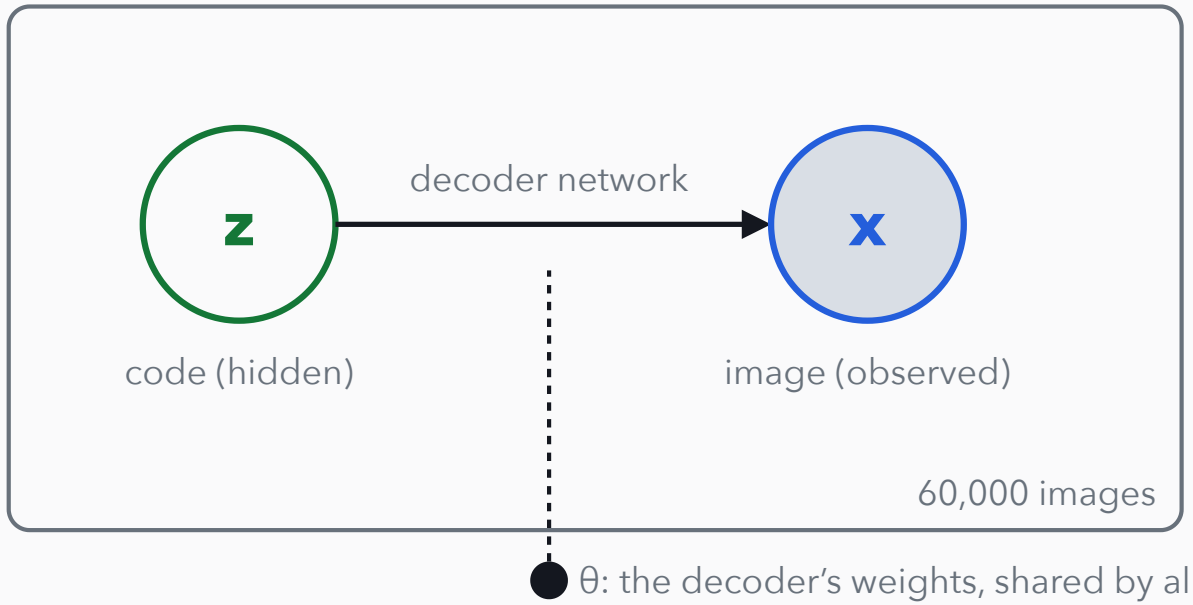
VAE samples are soft because each pixel is an independent Gaussian around a mean image.

Next: a generator trained against a second network that judges real from fake (GANs). Then many small denoising steps (diffusion).

> Lectures 3 and 4 replace it: a learned critic, then repeated denoising.

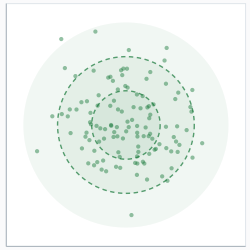
▶ Notebook

The VAE as a graphical model: a hidden code



the prior:

$p(z) = N(0, I)$, fixed in advance



the likelihood:

$p(x \mid z) = N(x; \text{decoder}(z), \sigma^2 I)$

each pixel: the decoder's value + noise of size σ

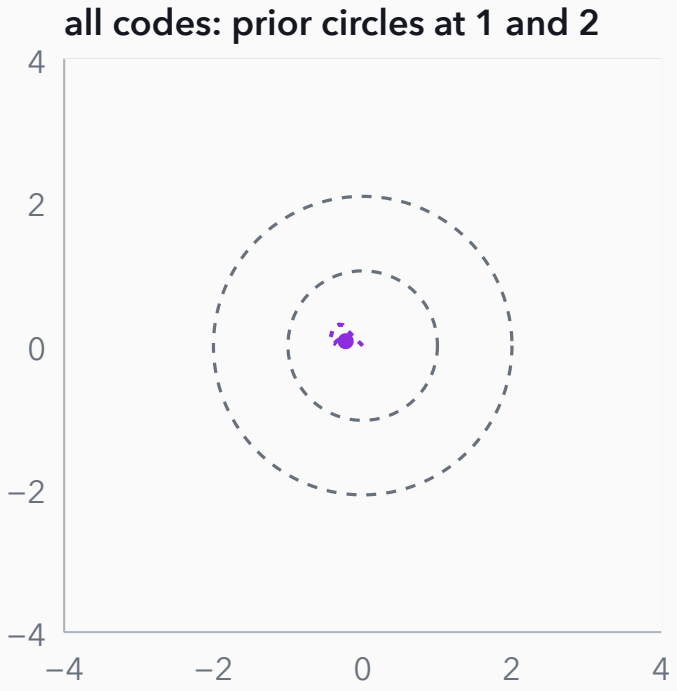
f(z), then x:

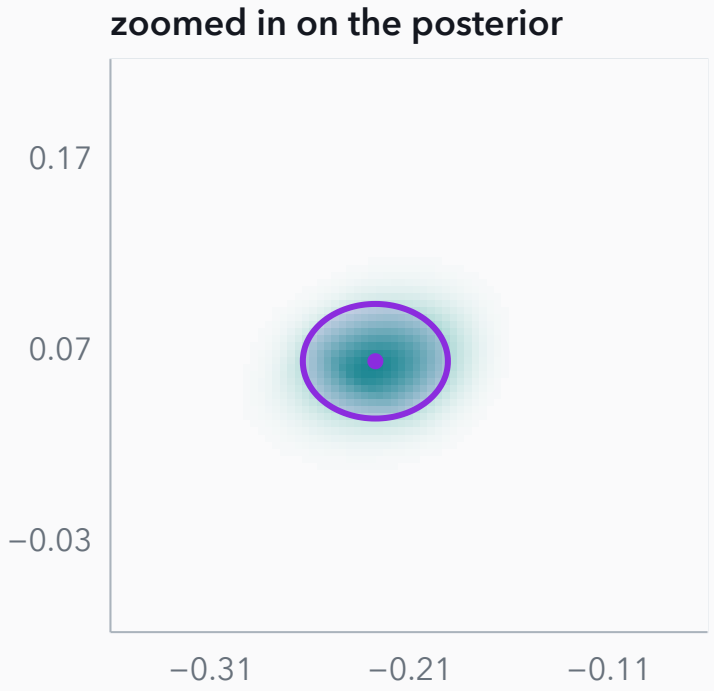


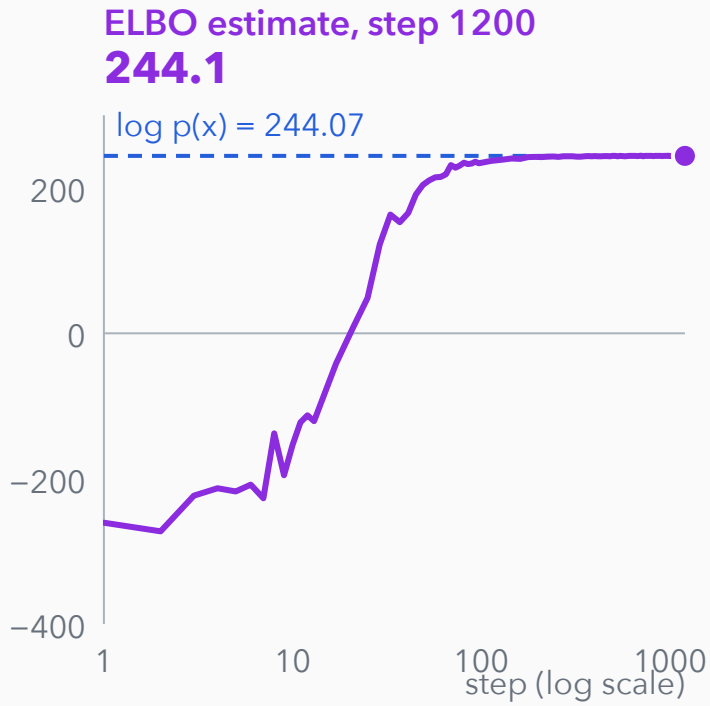
> The decoder's weights θ sit outside the box: one set, shared by every image, learned from the data.

▶ Notebook

Lecture 1's VI on one image: fit q for our 3, press play







> q moves into the posterior's patch and shrinks to fit it. The ELBO ends at 244.05, just below the ceiling log p(x) = 244.07.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

Did the autoencoder memorise? Training against test error

2-number code	training error	test error
autoencoder	30.3	31.1
PCA		43.6

Test digits were never used for training.

Training and test errors are close: it learned digits in general, not these particular 60,000.

> The gap is small (30.3 against 31.1), and the autoencoder beats PCA by a wide margin with the same 2 numbers.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

Without the ReLUs and the sigmoid, an autoencoder rediscovers PCA



Test error: linear autoencoder 43.8, PCA 43.6.

Its decoder spans almost the same plane as PCA's two directions (cosines of the angles: 1.000 and 0.92).

> Gradient descent found PCA's plane on its own: an autoencoder is PCA with bendable maps.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

The autoencoder: walk in a straight line from a 1 to a 0

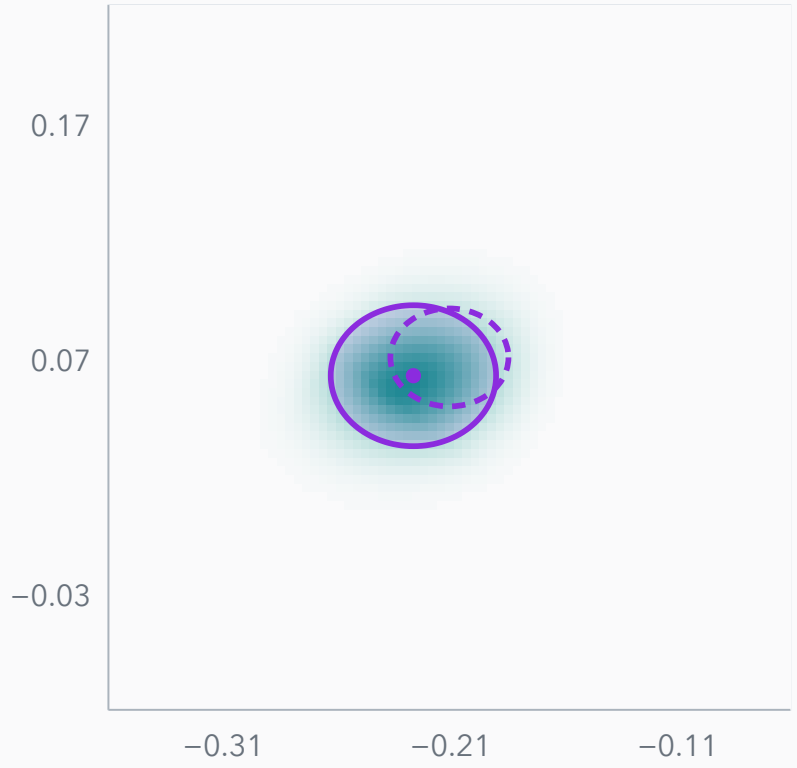


> With 2 numbers the path crosses the clusters of other digits; with 16 the middle passes through blobs that are no digit.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

Check the encoder: its q for our 3, against the true posterior



	value
log p(x), the exact target	244.07
ELBO, q fitted to this image alone	244.05
ELBO, q from the encoder (one pass)	243.37

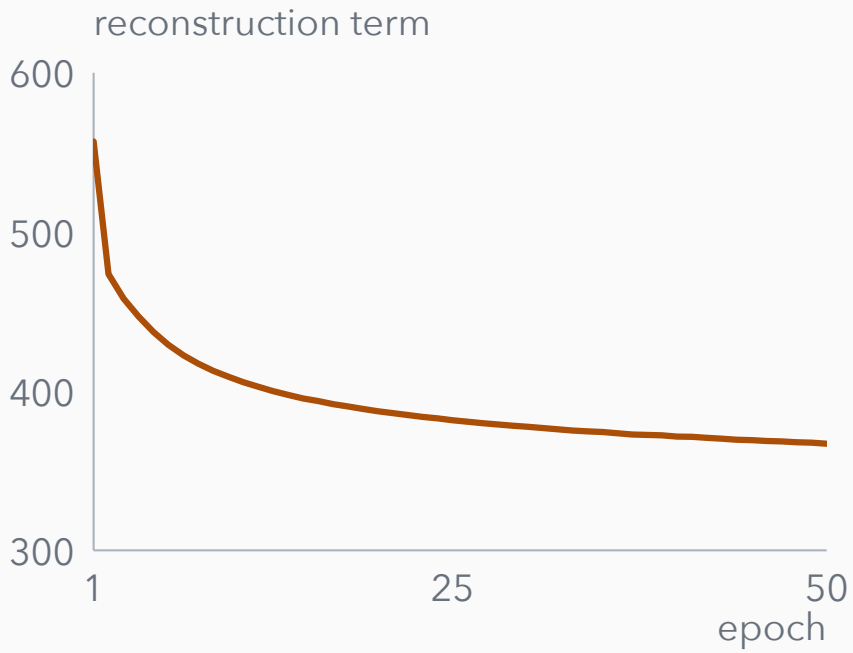
Dashed: the encoder's q. Shaded: the per-image fit. Both 2 standard deviations.

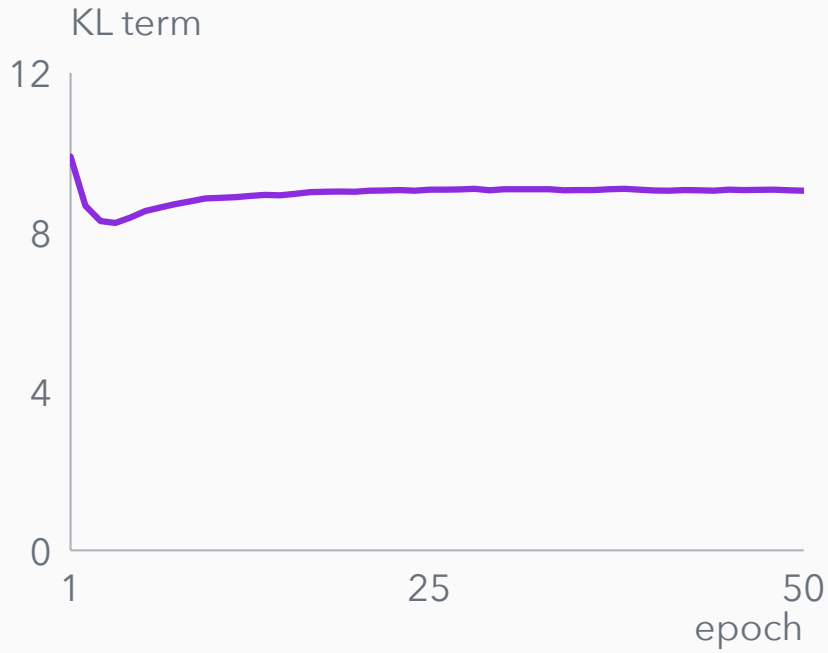
> The encoder's q sits slightly off-centre and a little too narrow: the price of one network serving all 60,000 images.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

While training: reconstruction falls, the KL settles





> After 50 epochs: reconstruction 367, KL 9.0 per image. The KL never falls to zero: then the code would say nothing about the image.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

β = 0: is it just an autoencoder?

β	KL per image	average σ of q	spread of codes
0	68.3	0.0001	5.36
1	9.0	0.0343	1.68
256	0.0	1.0001	0.00

With β = 0 nothing stops q from shrinking.

It still draws $z = \mu + \sigma\varepsilon$, unlike an autoencoder. But training drives σ to 0.0001, so it behaves like one.

> But training drives σ almost to zero, and then it behaves like one: spread-out codes and poor samples.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

β = 256: every code decodes to the same blur

β = 256, rebuilt



β = 256, samples



KL = 0.00: q(z | x) equals the prior for every image.

The decoder learns to ignore z and outputs the average digit. This is called **posterior collapse**.

> Then z carries no information, so the decoder ignores it and draws the average digit every time.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).